

A Comparative Study of Mental Health Detection Using PHQ-9 and Social Media Analysis with Traditional ML and NLP Techniques

Md Abdullah Ibne Aziz and Shahriar Ahmed

A Thesis in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

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Letter of Transmittal

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Subject: Letter of Transmittal.

Dear Sir,

It is our pleasure to present our thesis report entitled, “**A Comparative Study of Mental Health Detection Using PHQ-9 and Social Media Analysis with Traditional ML and NLP Techniques**” to your review and consideration. This thesis provides in-depth research on the detection of depression through automated methods with both structured clinical and unstructured social media texts. We honestly believe that this is the result of our personal work and academic integrity.

We believe that this thesis can fulfill the expectations of the department and, hopefully, make its contribution to the research in the field of machine learning-based mental health assessment. We would highly appreciate your precious comments and suggestions during evaluation. Thank you very much for your time and consideration.

Yours Sincerely,

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Student's Declaration

It is our pleasure to announce that the thesis report called “A Comparative Study of Mental Health Detection Using PHQ-9 and Social Media Analysis with Traditional ML and NLP Techniques” is the outcome of our original research work conducted under the guidance of Snaholata Mondal, Lecturer, Department of Computer Science and Engineering (CSE), International University of Business Agriculture and Technology (IUBAT).

This thesis has not been presented, or presented at all, to any other university or institution, to obtain any degree, diploma, or other academic qualification. Every information and materials provided in this thesis have been referenced appropriately by providing proper citation and reference points.

We also acknowledge that the work in this thesis is all our own unless due credit has been given and we have full responsibility for the authenticity and accuracy of the information we have in this thesis.

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Supervisor's Certification

This is to certify the fact that this thesis report was prepared and has been submitted executed by Md Abdullah Ibne Aziz (ID: 22203246) and Shahriar Ahmed (ID: 22203202) of IUBAT - International University of Business Agriculture and Technology in part fulfilment of the requirement of awarding Bachelor of Computer Science and Engineering (BCSE).

The thesis was conducted under my supervision and guidance, and the report displays the research work and tasks that had been completed by the students successfully. As far as I understand and in their own statement, no section of this thesis has been turned in previously to any other university or institution as part of a degree, diploma, or certification

The learners are allowed to present the thesis report to be reviewed. I hope they succeed in their academic and professional life.

Snaholata Mondal

Lecturer

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Abstract

Major Depressive Disorder is a typical mental health issue, which poses a serious health problem for the population. The thesis is dedicated to the issue of automated detection of depression through the comparison of structured clinical assessment data and unstructured textual data to interpret the data on depressive symptoms as one of the most prominent results of the Patient Health Questionnaire 9 (PHQ-9) survey, as well as to analyzing the depressive symptoms expressed in the text of the Bangladeshi university students on the Bengali social media. The research seeks to compare how well the traditional machine learning and natural language processing models can be used to screen individuals with mental health conditions and what are the advantages of each method. Machine learning models, such as Logistic Regression, Decision Tree, K-Nearest Neighbors, Random Forest, and XGBoost are applied to PHQ-9 data, whereas deep learning and transformer-based models, such as LSTM, multilingual BERT, and stacked ensemble framework are used on Bengali social media text. It is done through feature engineering, hyperparameter optimization, and stratified cross validation to achieve reliable performance and explainable artificial intelligence approaches, such as SHAP and LIME, to enhance interpretability. The experimental findings indicate that XGBoost model can obtain 96.35 percent accuracy and AUC ROC of 0.994 on PHQ-9 data, whereas NLP based ensemble model can obtain 95.78 percent accuracy on social media text. The results indicate that automated depression screening based on combining structured and unstructured textual analysis can be an efficient and scalable model to incorporate structured clinical data in the evaluation process.

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Chapter 1. Introduction

1.1 Introduction

Major Depressive Disorder (MDD) is a persistent and crippling mental ailment, which has a heavy toll on both individual and societal capacity in the global society in terms of emotional, cognitive, and physical functioning. Depression is also known to be among the most prominent causes of disability and diminished quality of life, which increases morbidity, mortality and economic loss in both developed and developing nations (Kroenke, Spitzer and Williams, 2001; Kim and Lee, 2021) [4][3]. According to recent estimates given across the globe, depressive disorders correlate to hundreds of billions of dollars each year in healthcare costs and productivity loss, and hence the dire need to have an effective and reproducible screening tool that would enable the detection and treatment of the condition at an early stage (Alves et al., 2025) [1].

In young adults, especially university students, depressive symptoms have become widespread significantly because of academic stress, financial burden, and psychosocial difficulties (Haq et al., 2022) [21]. In developing and developing nations like Bangladesh, access to mental health services is also limited and social stigma has continuously led to further diagnosis and treatment delays, further exacerbating the overall results of the condition (Dey et al., 2022) [10]. Through these issues, the need to establish automated, objective and available methods of screening depression is emphasized to supplement the traditional clinical evaluation.

One of the most resorted, clinically validated instruments in measuring the severity of depression is the Patient health questionnaire-9 (PHQ-9). It is a nine-item, self-administered questionnaire which uses DSM-IV criteria of diagnosis, and has shown high reliability, validity, and cross-cultural applicability in different populations (Kroenke, Spitzer and Williams, 2001; Lowe et al., 2004; Kroenke, 2021) [4][14][15]. Because of its conciseness and diagnostic efficiency, PHQ-9 is actively used in clinical practice and epidemiological studies (Inoue et al., 2012) [16].

Nevertheless, PHQ-9 is not always applied in everyday practice even though it has its strong aspects. Inadequate documentation, a low rate of administration and subjective self-reports reduce its effectiveness in longitudinal monitoring and outcome-based assessment (Ford et al., 2020; Gliklich et al., 2020; Ma et al., 2021) [17][18][19]. All these restrictions have encouraged the incorporation of machine learning methods to improve the predictive accuracy, coverage, and interpretability of PHQ-9-based screening mechanisms (Alves et al., 2025) [1].

1.2 Problem Statement

Even though PHQ-9 is a validated and commonly used tool of depression screening, its practical implementation is ineffective and susceptible to self-report bias, incompleteness in data capture, and inadequate longitudinal accessibility (Ford et al., 2020; Gliklich et al., 2020) [17][18]. Simultaneously, natural language processing-based solutions based on social media data provide scalable, passive monitoring solutions but do not necessarily have established clinical foundations and comparisons with established screening instruments (Skaik, 2021; Hossain et al., 2022) [24][10].

The general research is mostly divided between PHQ-9-based machine learning models and social media-based NLP models when it does not explicitly produce a direct and organized comparison within a single experimental framework (Glavin et al., 2022; Alves et al., 2025) [20][1]. This is especially notable in the case of Bengali language sample and Bangladeshi groups, which restricts the comprehension of the possibilities of using structured clinical measures and unstructured linguistic indicators to complete one another to achieve effective and scalable depression detection.

1.3 Objectives of the Study

The main aim of the study is to research and compare the automated depression detection methods with the use of both structured clinical questionnaire data and unstructured text in social media through machine learning and natural language processing methods. Specifically, the proposed study will assess the performance of traditional ML models, which are trained on PHQ-9 data, against NLP-based models which are trained on Bengali social media texts in terms of predictive performance, interpretability, and practical usefulness.

The final aims of the research are the following:

1. To create and test classical machine learning algorithms to detect depression based on the structured PHQ-9 questionnaire data of the Bangladeshi university students (Kroenke et al., 2001; Haq et al., 2022) [4][21].
2. To construct and apply NLP-based depression detection models on the Bengali social media text matched to PHQ-9 symptom dimensions (Hossain et al., 2022; Skaik, 2021) [10][24].

3. To evaluate the performance of PHQ-9-based ML models and social media-based NLP models based on the same set of evaluation metrics, including accuracy, precision, recall, F1-score, and AUC-ROC (Kim and Lee, 2021; Alves et al., 2025) [3][1].
4. To evaluate model interpretability with the help of explainable AI methods to determine the level of clinical transparency and decision-support appropriateness (Alves et al., 2025; Colledani et al., 2024) [1][9].
5. To investigate the viability of a bi-modal depression screening providing model that incorporate active clinical evaluation and passive linguistic surveillance.

1.4 Scope of the Study

This research focuses on automated detection of depression through two different and yet complementary data modalities, namely, the structured PHQ-9 questionnaire responses and the unstructured Bengali social media text. The PHQ-9 data utilized in the study was self-reported on Bangladeshi university students and consists of demographic items, PHQ-9 symptom assessment scores, and the choice of psychosocial variables in the academic background (Haq et al., 2022) [21].

In the structured data analysis, the research is interested in the traditional machine learning models, such as Logistic Regression, Decision Tree, K-Nearest Neighbors, Random Forest, and XGBoost, with the explainable AI approaches to interpretability, such as SHAP and LIME (Alves et al., 2025) [1]. The NLP-based model can process the Bengali-language social media text filtered and labeled based on PHQ-9 symptom dimensions and based on the

traditional ML, deep learning, and transformer-based architecture (Hossain et al., 2022; Razavi et al., 2020) [10][23].

The research is limited to dichotomous classification of depression to screen. This research will not cover clinical diagnosis, treatment recommendation, longitudinal outcome prediction, and analysis of multimodal physiological data. Also, the results are situation-specific, and they cannot be applied directly to non-students or other languages without additional verification.

1.5 Contribution of the Study

This study has several significant implications on automated mental health assessment and depression screening research.

First, it provides a direct and systematic comparison of machine learning models based on PHQ-9 and social media-based NLP models within a single experimental setting, which has not been previously studied in the literature because these two approaches are usually considered independently (Glavin et al., 2022; Alves et al., 2025) [20][1].

Second, the research has the contribution of low-resource language literature in creating and testing NLP-based models of depression detection in Bengali social media text, which has not been studied extensively despite the number of Bengali speakers and the rise in online interaction (Hossain et al., 2022) [10].

Third, explainable AI methods increase both model transparency and clinical interpretability, which promotes the possibility of using machine learning systems as decision-support systems, and not black-box opaque models (Colledani et al., 2024; Alves et al., 2025) [9][1].

Lastly, this paper suggests a dual-modality viewpoint, which emphasizes the strengths of structured clinical screening instruments and passive linguistic surveillance and suggests that this presents a scalable and ethically viable method of detecting early depression and mental health monitoring of the population at large.

Chapter 2. Literature Review

2.1 Introduction

Automated depression detection literature has been growing at a significant pace in the past two decades due to the increasing burden of the major depression disorder (MDD) in the world and the development of machine learning and natural language processing technologies. Current studies can be broadened into four main groups namely: validated clinical screening measures of depression, machine learning methods based on the structured questionnaire data, depression detection based on unstructured digital information by use of NLP and comparative/integrative models of different modalities.

2.2 Clinical Screening Instruments for Depression

Depression assessment has its basis on validated clinical questionnaires, which are considered the gold standard of diagnosis, as well as screening. The Patient Health Questionnaire-9 (PHQ-9) is the most common in measuring the severity of depressive symptoms. The PHQ-9 is a short, self-administered questionnaire developed by Kroenke, Spitzer and Williams (2001) based on the DSM-IV criteria of diagnosis, and showed a high level of internal consistency, sensitivity and specificity in clinical populations [4]. Its effectiveness in the process of monitoring of treatment outcomes along with longitudinal change of symptoms was later supported by other studies (Lowe et al., 2004) [14].

Philosophy PHQ-9 has been widely embraced in the world, and Kroenke (2021) refers to its cross-cultural reliability and suitability in various healthcare systems [15]. Inoue et al.

(2012) also confirmed the PHQ-9 in clinical psychiatric settings, which does not only have a viable use, but also assessment instruments administered by clinicians [16]. All these studies prove that PHQ-9 is a powerful and clinically valid screening instrument.

Although this is a good study, various constraints have been indicated in practice. Ford et al. (2020) have reported discrepancies in the implementation of the PHQ-9 when it was applied in patient-physician encounters and was usually affected by a time limit and clinical workload [17]. Equally, incomplete documentation and unreliability of the outcome measures were highlighted as obstacles to effective longitudinal monitoring by Gliklich et al. (2020) [18]. Ma et al. (2021) have shown inconsistencies between self-reported PHQ-9 and observed severity of depression, which is dependant on personality traits and dysfunctional attitudes contributing to self-report bias [19]. These results indicate that PHQ-9 is a valid clinical measure, but the effectiveness of this tool can be increased with algorithmic and objective methods of analysis.

2.3 Machine Learning Approaches Using PHQ-9 Data

To address the constraints linked to manual coding and subjectivity, scholars have continued to use machine learning methods on coded PHQ-9. Prior studies have shown that conventional ML classifiers would succeed in forecasting depression status based on responses to PHQ-9 and demographic factors. According to Vidya, Swamy and Lalitha (2024), random forest and XGBoost showed even greater accuracy in depression prediction compared to linear models [2], making them ensemble-based models.

A comparative analysis of the ML algorithms based on PHQ-9 and GAD-7 data was conducted by Haq et al. on the population of Bangladeshi university students, with the highest accuracy of prediction of depression obtained with Random Forest and the best performance of Support Vector Machines with anxiety [21]. These results support the appropriateness of tree-based and ensemble techniques in the non-linear representation of the relationships between depressive symptoms.

More sophisticated methods have been based on optimization of the PHQ-9 items. Glavin et al. (2022) and Glavin et al. (2023) proved that certain combinations of PHQ-9 items may be more efficient than conventional ultra-brief screening instruments like PHQ-2, indicating that data-guided item selection can help to improve screening performance without compromising any accuracy [20]. Colledani et al. (2024) also demonstrated that personalized assessment and interpretation were achieved using Decision Tree models using demographic variables with PHQ-9 items [9].

Another major step was made by Alves et al. (2025), who suggested an ML-based framework (ePHQ-9), which combines structured PHQ-9 data with unstructured clinical notes on electronic medical records [1]. Their model of ensemble was able to produce large AUC values and show the possibility of automated estimation of symptom severity at scale. Notably, this study has focused on the explainability of its work, which was achieved by analyzing the importance of features, which aligned with clinical reasoning and prediction by ML.

2.4 Explainable AI in Depression Detection

With the increasing complexity of machine learning models, the issues of transparency and clinical trust have resulted in the integration of explainable machine learning (XAI). The issue of explainability is especially urgent when used in mental health, where outcomes of decisions could have an impact on patient outcomes and clinical intervention.

The most frequently used XAI in depression detection studies have become SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). Alves et al. (2025) demonstrated that SHAP values could release the relative usefulness of individual PHQ-9 items to forecast the extent of the depression, therefore permitting clinicians to analyze the model outputs in a symptom-focused manner [1]. The study by Colledani et al. (2024) also demonstrated that interpretable tree-based models demonstrated a better acceptance among clinicians and allowed finding personalized diagnostic insights [9].

These articles demonstrate that explainability is effective in enhancing transparency and reducing the disparity between data-driven models and clinical decisions, making it morally acceptable to use AI in mental health care.

2.5 NLP-Based Depression Detection Using Social Media Data

Simultaneously with questionnaire-based methods, NLP passive monitoring of depression and continuous monitoring NLP-based analysis of social media text has seen increased popularity as an approach to depression monitoring. The social media platforms

enable people to share emotions, thoughts, and experiences using natural language and as a result, these make them useful in identifying psychological distress.

As Skaik (2021) showed, population-level predictions of depression and suicidal ideation with the help of linguistic indicators of such behavior based on social media posts were possible [24]. In the above application, Hossain et al. (2022) used Linguistic Inquiry and Word Count (LIWC) features with the Random Forest classifiers on Facebook data and obtained positive results of the identification of depressive language patterns [10]. Their results revealed that the first-person pronouns, negative emotion words and absolutist words were powerful indicators of depression, which agrees with previous psycholinguistic theories.

Later research has used deep learning designs to enhance performance. Razavi, Gharipour and Gharipour (2020) took advantage of smartphone and social media metadata and incorporated the data into the ML model that identifies cases of depression, and the importance of passive behavioral data cannot be overstated [23]. Choudhary et al. (2022) also showed that the models based on digital behavioral patterns, validated on PHQ-9 scores, could be useful in classifying depression with help of the Random Forest models [22].

NLP-based depression detection has been improved tremendously with the help of transformer-based models like BERT. Kim and Lee (2021) found significant pooled AUC values of depression screening by MLs using PHQ-9-validated digital data sources, which points to the usefulness of contextual language models [3]. Nevertheless, the majority of

current studies in the field of NLP are based on datasets in English language, which undermines low-resource languages, including Bengali.

2.6 Research Gap and Motivation

Even though there is plenty of research on PHQ-9-based machine learning models and social media-based NLP methods, both fields are researched almost separately. Most research works do not consider PHQ-9 as a modeling domain but only as a validator of digital behavior study (Choudhary et al., 2022; Razavi et al., 2020) [22][23]. On the other hand, NLP-aligned research tends to have no explicit comparison with clinically tested screening tools, which inhibits the ability to interpret and embrace them in medical care (Skaik, 2021) [24].

Moreover, the lack of studies about the data on Bengali language and the Bengali populations, although the case of depression is prevalent among the university students in this area (Haque et al., 2022) [21], is quite significant. A systematic comparison framework that would compare an organized PHQ-9-based ML model and an unstructured social media-based NLP model on similar experimental terms does not exist in literature.

2.7 Summary

In conclusion, the literature review confirms PHQ-9 as a clinically sound screening instrument and proves the efficiency of machine learning models as methods of improving the detection of depression based on the information provided in the form of structured questionnaires. At the same time, NLP-oriented solutions with social media text provide in-scale and passive solutions to mental health monitoring. Nevertheless, there are no direct

comparative studies, scant emphasis on low-resource languages, and the absence of incorporating explainable AI methods, which indicate evident knowledge gaps in research. The strategy of filling these gaps is the main impetus of the current research.

Chapter 3. Research Methodology

This paper introduces a clinically sound and methodologically sound machine learning model of automated depression diagnosis based on the Patient Health Questionnaire-9 (PHQ-9). We selectively use primary data that has been gathered by us and in this way, we can have complete control on the survey design, population attributes and quality of data. Through the combination of proven psychometric tools and conventional machine learning methods, we would like to research how patterns of depressive symptoms and stressful conditions in the specific context can be used to conduct a valid depression screening of university students.

3.1 PHQ-9 Based Analysis

3.1.1 Dataset Description

The PHQ-9 Student Depression Dataset which we built and maintained was created in a structured self-report survey given to Bangladeshi university students. The research process was undertaken following the accepted ethical standards in research such as informed consent, voluntary participation, and anonymity of all the personal identifiers. We also published the dataset publicly on Mendeley Data [26] to achieve a degree of transparency and reproducibility.

All records in the dataset are individual participants and include demographic information, in this case age and gender, as well as a response to questions in the PHQ-9 which are a total of nine. These scales reflect depressive symptoms in the last two weeks and are on the emotional, cognitive, somatic, and risk-related aspects of depression. The PHQ-9 questions evaluate the loss of interest or pleasure, the continued low mood, sleeping disturbances,

fatigue, change in appetite, the feeling of worthlessness, difficulty focusing, psychomotor changes, and thoughts of self-harm.

Besides the fundamental PHQ-9 questions, we also added contextual psychosocial variables in the context of the university setting that are specifically salient. They are self-reported quality of sleep, perceived study or academic pressure and financial pressure. These variables combined, we hoped to be able to model depression as not only a profile of clinical symptoms but also as a process that is affected by lifestyle and environmental stressor factors.

Table 3.1: Summary of Dataset Variables and Descriptions

Feature Category	Variable Name	Description	Data Type
Demographic	Age	Age of participant in years	Numerical
Demographic	Gender	Self-reported gender	Categorical
PHQ-9 Item	Interest_Pleasure	Loss of interest or pleasure	Ordinal
PHQ-9 Item	Feeling_Down	Depressed mood	Ordinal
PHQ-9 Item	Sleep_Trouble	Sleep disturbance	Ordinal
PHQ-9 Item	Tired_Low_Energy	Fatigue or low energy	Ordinal
PHQ-9 Item	Appetite_Issues	Appetite or eating changes	Ordinal
PHQ-9 Item	Feeling_Bad_About_Self	Feelings of worthlessness	Ordinal

PHQ-9 Item	Concentration_Trouble	Difficulty concentrating	Ordinal
PHQ-9 Item	Moving_Speaking_Issues	Psychomotor changes	Ordinal
PHQ-9 Item	Thoughts_Self_Harm	Thoughts of self-harm	Ordinal
Psychosocial	Sleep_Quality	Perceived sleep quality	Ordinal
Psychosocial	Study_Pressure	Academic stress level	Ordinal
Psychosocial	Financial_Pressure	Financial stress level	Ordinal

3.1.2 Data Preprocessing

The data preprocessing pipeline before model development gave the results consistency, interpretation, and algorithmic appropriateness. To simplify the process and ensure consistency of the findings, we standardized the names of columns in line with the constructs that are clinically meaningful and consistent terminologies of PHQ-9. This measure helped in easier interpretation of features both in the process of modeling and analysis.

All categorical and ordinal data, such as gender and individual PHQ-9 responses, PHQ-9 severity labels and psychosocial stress indicators, were converted into numerical forms through label encoding. This method maintained the natural ordinal associations of symptom severity scales as well as compatibility with conventional machine learning algorithms.

We also checked the data on missing values, outlier, and distributional anomalies. The descriptive statistics were calculated to provide an overview of the central tendencies and variability of the key features.

3.1.3 Feature Engineering

Our massive feature engineering was done to improve predictive performance and clinical interpretability. A binary target variable was initially created where we defined clinically significant depression. Based on accepted PHQ-9 criteria, we identified those participants who had a total PHQ-9 score of ten or above as depressed. This formulation allowed us to model the problem of depression detection as a supervised binary classification problem.

In addition to the binary result, we combined separate PHQ-9 items into symptom domains that are clinically rational. Core emotional symptoms were the sum of scores of diminished interest and depressed mood. Physical symptoms were obtained based on sleeping disturbance, fatigue, and appetite variation and cognitive concentration problems and negative self-perception. The symptoms related to risks were measured with the help of psychomotor changes and self-harm ideation. We also calculated a total symptom score, which is a summation of domain scores, a general measure of the depressive burden.

We also designed explicit risk indicators, which can be used in clinical risk stratification. We established a high-risk depression measure of those individuals with PHQ-9 scores that corresponded to moderately severe or severe depression. We also developed a suicidal risk predictor which is high scores on the self-harm item, and we acknowledge that it is very important in mental health assessment. These are the engineered features, and their clinical interpretations as summarized in Table 3.2.

Table 3.2: Target Variable and Symptom Grouping

Feature Name	Definition	Clinical Significance
Depression_Binary	PHQ_Total ≥ 10	Clinically significant depression
Core_Symptoms	Q1 + Q2	Emotional symptom severity
Physical_Symptoms	Q3 + Q4 + Q5	Somatic symptom burden
Cognitive_Symptoms	Q6 + Q7	Cognitive impairment
Risk_Symptoms	Q8 + Q9	Behavioral and suicidal risk
High_Risk	PHQ_Total ≥ 15	Moderately severe to severe depression
Suicidal_Risk	Q9 ≥ 2	Elevated suicide ideation risk

3.1.4 Proposed Models

- Random Forest

Dataset Representation:

$$D = \{(x_i, y_i)\}_{i=1}^n, x \in \mathbb{R}$$

Each tree h_t is trained on bootstrap samples. Final prediction uses majority voting:

$$\hat{y} = \text{mode}\{h_t(x)\}_{t=1}^T$$

RF is repetitive and produces a group of trees from the random input and features; The main “class” that show the greatest output, which reduces overfitting and variance.

- Decision Tree

$$G(p) = 1 - \sum_{c=1}^c p_c^2$$

Quantifies how pure the classes at a node are; splits should be made to minimize impurity and create better separate classes.

- Logistic Regression

Binary Classification probability:

$$z = w^T x + b$$

$$P(y = 1|x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Multiclass (Softmax):

$$P(y = c|x) = \frac{e^{w_c^T x}}{\sum_{j=1}^c e^{w_j^T x}}$$

An additional class probability sigmoid(binary) or softmax(multiclass) is applied to a linear function of outputs.

- K-Nearest Neighbors (KNN)

Distance-based similarity:

$$d(x, x_i) = \sqrt{\sum_{j=1}^d (x_j - x_{ij})^2}$$

The class is estimated by the nearest K samples in the training set with Euclidean distance and majority rule.

- XGBoost

Additive tree model:

$$\hat{y}_i = \sum_{t=1}^T f_t(x_i), \quad f_t \in \mathcal{F}$$

Regularized objective:

$$\mathcal{L} = \sum_i l(y_i, \hat{y}_i) + \sum_t \Omega(f_t)$$

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$$

Every tree fixes the mistakes that have been made so far (boosting). The regularization term can control the complexity of the model and avoid overfitting.

- SHAP

Feature importance via Shapley values:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

Measures the importance of a feature by looking at the differences in model results when a feature is present or not for all combinations.

- LIME

Local surrogate explanation:

$$g = \arg \min_{g \in G} [\mathcal{L}(f, g, \pi_x) + \Omega(g)]$$

Approximates the original model locally near a particular example by a simpler interpretable model, to explain why a prediction was made.

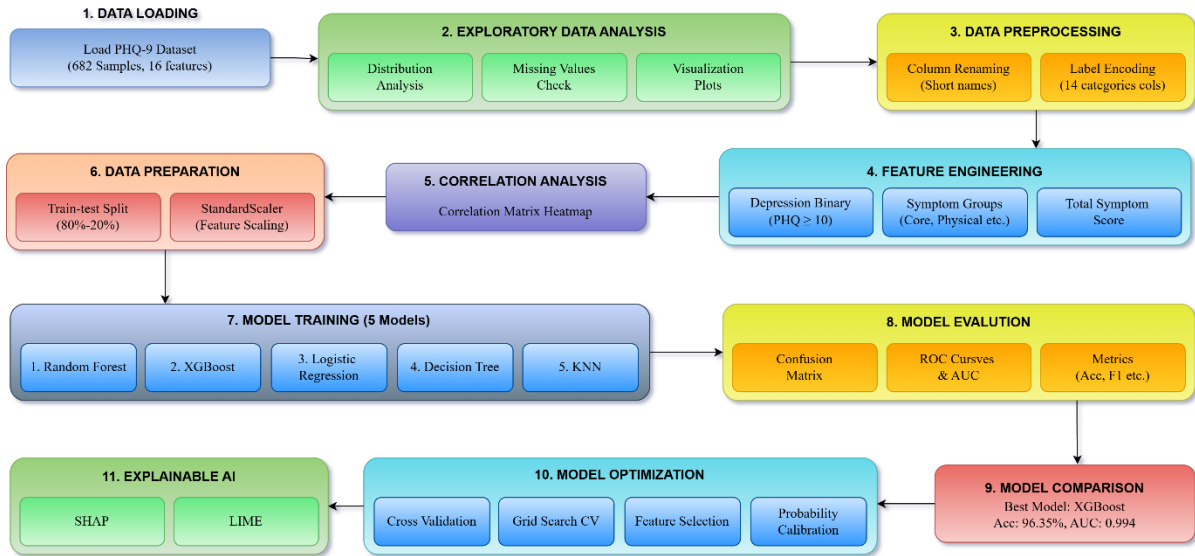


Figure 3.1 PHQ-9 Methodology (Traditional ML) Pipeline

3.1.5 Training Pipeline

All the models were trained with a common and reproducible pipeline. We divided the dataset into a training and testing dataset in an 80: 20 stratified split to maintain the original class distribution. To make sure that the magnitude-sensitive models did work well, we used standard normalization methods to scale the features. To increase the level of robustness and minimize variation in performance estimation, we used ten-fold stratified cross-validation in training.

3.1.6. Hyperparameter Optimization

We conducted hyperparameter optimization based on grid-based search strategies and specifically on the XGBoost classifier. We systematically searched over combinations of parameters that govern the model complexity, learning rate and subsampling behavior in the cross-validation framework. This was done to make sure that performance increases obtained were as a result of actual learning and not due to overfitting certain data partitions.

3.1.7 Feature Selection

We used a variety of feature selection methods to select the most informative predictors. The feature importance scores obtained using the trained XGBoost model were used to evaluate relative importance. We used Recursive Feature Elimination to narrow down the set of features, thereby eliminating irrelevant features, and we calculated the mutual information scores to get the linear and non-linear associations between the features and the outcome of the depression. These methods were converged, which strengthened the powers and clinical plausibility of the predictors chosen.

3.1.8. Probability Calibration

Since there is a clinical implication of the screening of depression, we tuned the predicted probabilities to make sure that the risk is estimated. Isotonic regression and sigmoid-based Platt scaling were the post-hoc methods of calibration we used. These techniques matched the predicted probabilities with observed outcome frequencies, which enhanced interpretability, and were also consistent with possible implementation in decision-support systems.

3.2 PHQ-9 Based Analysis (Natural Language Processing)

Besides structured questionnaire-based modelling, we also extend our methodological framework on the unstructured textual data to elicit linguistic manifestations of depressive

symptoms. We consider that people intuitively express emotional distress more using language than using formal questionnaires and hence we perform a text analysis of Bengali social media with deep learning-based natural language processing methods. This element is complementary to PHQ-9 structured analysis because it allows one to identify depression based on ad hoc and real-world textual expressions, which increases ecological validity and practical usefulness.

3.2.1 Dataset Description

In the case of the NLP-based analysis, we refer to self-curated data that we built to analyze depression in Bengali language. The dataset is composed of 2, 841 Bengali sentences, which were gathered on the social media-like situations and were manually correlated on the depressive symptom categories. Every sentence constitutes a brief textual manifestation that could be emotional, cognitive, physical or risk related to depression.

The data set includes three main columns, including category, sentence, and the level of severity. The category column is used to assign each sentence to one of the PHQ-9 symptom dimensions, which conceptually aligns the two methods of analyzing the data, the questionnaire-based and the text-based. Raw Bengali text is written in the sentence column, with occasional English words, numerals, emojis, and informal symbols, which is impersonal to the natural language behavior of online communication. The level of perceived severity column is coded into the perceived severity of the depressive expression in relation to each sentence.

To categorize the problem, we record the severity levels into two categories (Depressed and Not Depressed) and make it a binary problem. Such binarization allows making a direct

comparison with the binary depression outcome as employed in the classic analysis of machine learning.

Table 3.3: Overview of the Bengali NLP Dataset

Attribute	Description
Language	Bengali (with code-mixed English)
Total Samples	2,841 sentences
Text Length	Short-form social media expressions
Categories	Aligned with PHQ-9 symptom dimensions
Labels	Depressed, Not Depressed
Annotation	Manually curated and severity-aligned

3.2.2 Text Preprocessing

Considering the informal and noisy characteristics of social media texts, we have constructed an elaborate preprocessing pipeline to standardize linguistic differences without losing semantically meaningful text. We have applied our own text-cleaning algorithm to Bengali mixed text of English. This feature transforms all characters into lowercase, excludes URLs and user mentions and hashtags, and removes mathematical characters and punctuation characters. We also delete emojis, special symbols, and non-linguistic artifacts and clearly maintain the semantic integrity by keeping the Bengali and English alphabetic characters.

To handle lexical noise, we do whitespace normalization and use a combined stopwords repository to do stopwords removal. We create a tailor-made, all-encompassing Bengali

stopword list, edited by linguistic examination and frequency check of the word list in the corpus, and combine it with the standard English stopwords list available in the NLTK library. The rationale behind this bilingual stopwords approach is that the mixed language in Bengali social media text is common.

Its preprocessing pipeline guarantees the resulting text to be linguistically clean and semantically focused as well as appropriate to the transformer-based tokenization. On Table number 4, a summary of preprocessing steps and their goals are given.

Table 3.4: Summary of Preprocessing Steps

Step	Purpose
Lowercasing	Reduce lexical variability
URL and Mention Removal	Eliminate non-semantic tokens
Punctuation and Number Removal	Reduce noise
Emoji and Symbol Removal	Retain only linguistic content
Stopword Removal (BN + EN)	Focus on semantically meaningful words
Whitespace Normalization	Improve tokenization consistency

3.2.3 Proposed Model

- LSTM

Sequential memory and gating:

$$f_t = \sigma(W_f[h_{t-1}, x_t])$$

$$i_t = \sigma(W_i[h_{t-1}, x_t])$$

$$C_t = f_t \odot C_t - 1 + i_t \odot \tilde{C}_t$$

Using gates to decide which information should be thrown away or kept and passed through, allowing gradients to flow for long durations.

- mBERT

Self-attention mechanism:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Learns the context relationship by contrasting each word with all other words using higher statistics for relevant words.

- Stacked Ensemble

Meta-feature vector:

$$z = [\mathcal{PRF}, \mathcal{PLR}, \mathcal{PKNN}, \mathcal{PLSTM}, \mathcal{PmBERT}]$$

Final Prediction:

$$\hat{y} = \arg \max(softmax(Wz + b))$$

It aggregates predictions of all models; the meta-classifier will learn how reliable each model is for making final decisions.

3.2.4 Training Configuration

For an exhaustive scope for depression detection in Bengali social media text, we developed multiple NLP models based on traditional MLs, RNNs and transformer-based approaches. This multi-model approach allowed us to systematically explore the trade-offs between representational capacity, computational complexity, and performance on low-resource Bengali text data.

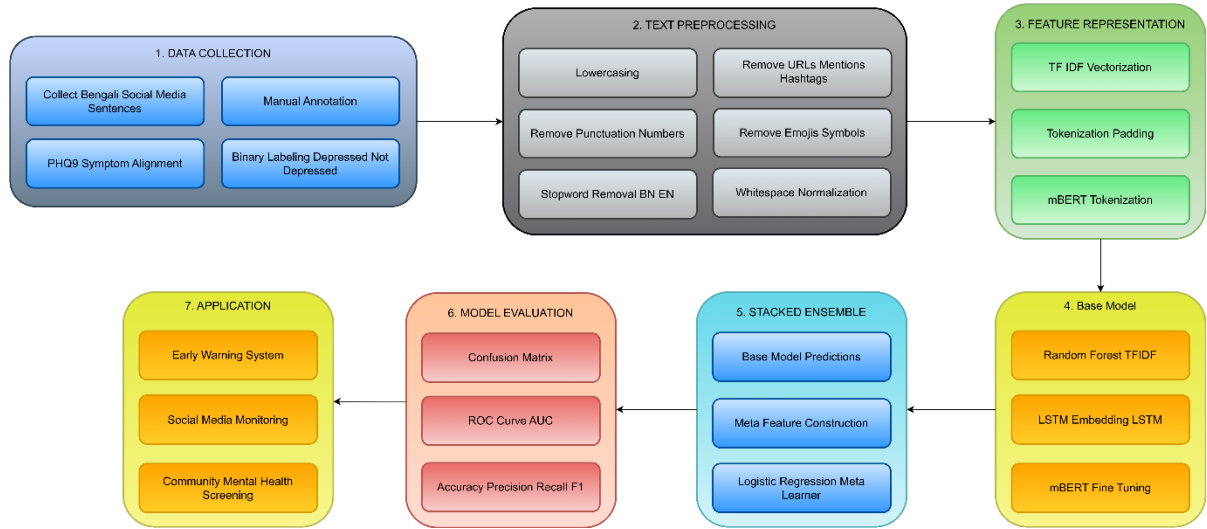


Figure 3.2 PHQ-9 Methodology (NLP) Pipeline

3.2.5 Traditional Machine Learning Configuration

As baseline, we trained a Random Forest classifier with some vectorized textual representations. We converted preprocessed sentences to numerical feature vectors based on the term-frequency-inverse document frequency (TF-IDF) encoding. This model extracts word-level salience in automatic and computationally efficient manner.

We developed the Random Forest model based on an ensemble of decision trees and aimed to learn robustly from high-dimensional sparse textual feature. The hyperparameters (estimators and max depth of tree) were chosen empirically between bias versus variance. The dataset was divided into training and validation sets in an 80:20 proportion while

maintaining the class distribution from the original dataset. This would be a strong non-deep-learning benchmark model to compare more advanced architectures.

A summary of the traditional machine learning configuration is presented in Table 3.5.

Table 3.5: Training Configuration for Random Forest Text Classifier

Parameter	Value
Feature Representation	TF-IDF
Classifier	Random Forest
Train-Validation Split	80:20
Task	Binary classification
Language	Bengali (code-mixed)

3.2.6 LSTM-Based Deep Learning Configuration

To capture sequential dependencies and word-ordering information, we utilised LSTM network to perform the depression classification. We tokenized and converted preprocessed text to human readable integers. Fixed length padding of these sequences was applied.

The LSTMs model architecture included an embedding layer to map words into dense vectors of real numbers and one or more LSTM layers to learn temporal dependencies in texts. A fully connected output layer projected the learned sequence representation into binary class probabilities. We used dropout regularization to reduce overfitting, since the number of samples in the dataset is limited.

We used categorical cross-entropy loss and optimized the LSTM model with gradient-based optimization technique. The dataset was partitioned into training and validation data with a ratio of 80:20, in accordance with other models to make a fair comparison. Previous studies have used such architecture to capture language patterns that could not be learned by bag-of-words models.

Table 3.6: Training Configuration for LSTM-Based Classifier

Parameter	Value
Model Type	LSTM
Input Representation	Tokenized sequences
Padding Strategy	Fixed-length padding
Loss Function	Cross-Entropy
Regularization	Dropout
Train-Validation Split	80:20
Task	Binary classification

3.2.7 BERT-Based Transformer Configuration

For knowledge-aware language understanding, we fine-tune a BERT-based transformer model on bert-base-multilingual-cased pretrained checkpoint for CLEN propagated information. This architecture can perform unsupervised and draft supervised pre-training for analyzing Bengali. We tokenized all sentences using the appropriate tokenizer and limited sentence length to 64 tokens.

We trained the model end-to-end with a batch size of 8 and learning rate of 2×10^{-5} - proven to be stable for transformer training on small to medium sized datasets. To facilitate appropriate regularization, we used AdamW optimizer with weight decay. The loss function was categorical cross-entropy, and it trained for 10 epochs. Results were consistent across models after omitting approximately 5% data for the train-validation split (80:20).

Table 3.7: Training Configuration for BERT-Based Classifier

Parameter	Value
Pretrained Model	bert-base-multilingual-cased
Maximum Sequence Length	64
Batch Size	8
Learning Rate	$2e-5$
Optimizer	AdamW
Loss Function	CrossEntropyLoss
Epochs	10
Train-Validation Split	80:20
Task	Binary classification

3.2.8 Methodological Justification

By using RF, LSTM and BERT models, we cover a wide range of representational paradigms from interpretable feature-based to sequential deep learning to contextualized transformers. By layering this experimental design, we can measure performance

improvements from increasing model complexity in a way that is transparent and reproducible.

3.3 Experimental Setup

In this section, we explain the experimental setting, software infrastructure, evaluation conditions, and measures of reproducibility of the experiments performed in all the experiments. The system was created to provide methodological uniformity, even comparison between the models and reproducible results both to structured PHQ-9 data analysis and unstructured Bengali text-based modeling.

3.3.1 Hardware Configuration

All experiments were conducted on a computer system with CUDA and NVIDIA GTX 1660 Super. Deep learning models, e.g. LSTM and BERT, were trained and inferred by using the acceleration of the GPU, whereas the machine learning models, i.e. Logistic Regression, Random Forest, K-Nearest Neighbors, Decision Tree, and XGBoost could be trained and run by the CPU on the same runtime environment. The system was also capable of storing big feature matrices and tokenized text data and the presence of GPUs was checked before training to take proper CUDA usage.

3.3.2 Software Stack

We used version Python 3.13 to implement all experiments, and they were compatible with current machine learning and deep learning libraries. To perform classic machine learning tasks with structured PHQ-9 data, we have used the scikit-learn ecosystem to implement models, preprocess in the model, cross-validate and evaluate the models. Gradient-boosted tree modeling and feature importance analysis were done with the help of XGBoost.

PyTorch served as the fundamental neural network as text classification models based on deep learning were deployed. To model on transformers, we used the Hugging Face Transformers library to load and fine-tune the multilingual model of BERT. NLTK was also utilized to process text, tokenize it and remove stopwords whereas pandas and NumPy were heavily employed to manipulate data and perform numerical calculations.

To analyze the visualization, performance and results inspection, we used matplotlib and seaborn to produce plots about the model performance, feature importance, and calibration behavior. The unified software stack was uniformly applied on all three notebooks to achieve the coherence of the experiments.

3.3.3 Evaluation Metrics

To assess the performance of the models, we considered a set of metrics that are suitable in binary classification of depression. To evaluate the accuracy of the classification, we calculated the accuracy, precision, recall, and F1-score, which allowed the balanced assessment of the accuracy depending on the distribution of classes and sensitivity to the depressive cases. In order to estimate the discriminative capacity of probabilistic classifiers we also calculated area under the receiver operating characteristic curve (AUC-ROC).

Since the estimates of probability are clinically relevant in depression screening, we did not only evaluate them in terms of classification, but also in terms of calibration. To measure the compatibility between predicted probabilities and the observed results, we calculated the Brier score especially when the probability model used probability calibration methods like isotonic regression and sigmoid scaling.

In the case of models trained on cross-validation, especially when using the classic machine learning pipelines, we provide the average and the standard deviation of all the results of the

evaluation metrics between folds. The method gives a strong approximation of the generalization performance and minimizes the dependence on the train-test split.

Chapter 4. Results and Discussion

Here we introduce and provide a critical analysis of the experimental results of the two parts of our proposed framework namely traditional machine learning models of PHQ-9 survey and natural language processing models of Bengali social media. We highlight not just the predictive performance but also the strength, decipherability, calibration and clinical importance. Combining quantitative assessment with explainable AI methods, we hope to achieve a detailed evaluation of the practicality of the suggested methods of depression screening.

4.1 Result of PHQ-9 Survey Based Models (Traditional ML)

4.1.1 Dataset Characteristics and Class Distribution

The PHQ-9 dataset had 682 samples which were gathered among the students at the university, and 16 original variables, such as demographics and nine PHQ-9 questionnaire questions. We systematically engineered the feature space to 24 variables by introducing the feature groupings and risk indicators. The binary target variable was characterized by quite equal distribution, 52.93 per cent samples were non-depressed and 47.07 per cent were depressed. The balance is especially beneficial to the supervised learning, since it avoids bias in the majority class, and metrics of evaluation are free to capture the true discriminative capacity.

Table 4.1: Summary of the Dataset Composition and Train-Test Split

Characteristic	Value
Total Samples	682
Original Features	16
Engineered Features	24
Training Set	545 (80%)
Test Set	137 (20%)
Depressed Class	321 (47.07%)
Non-Depressed Class	361 (52.93%)

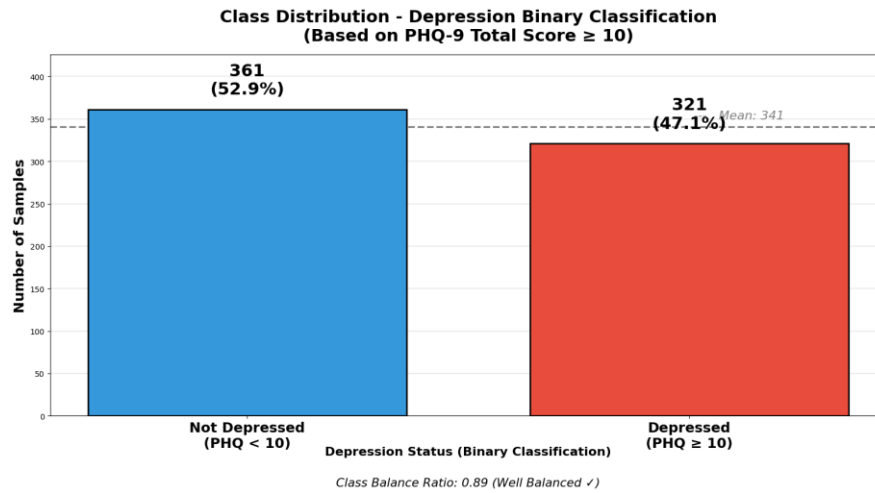


Figure 4.1: Class Distribution (Depressed vs Not Depressed)

4.1.2 Model Performance Comparison

Table 4.2: Classification Table of Random Forest Classifier

Class	Precession	Recall	F1-Score	Accuracy
Not Depressed	0.89	0.90	0.90	0.8905
Depressed	0.89	0.88	0.88	

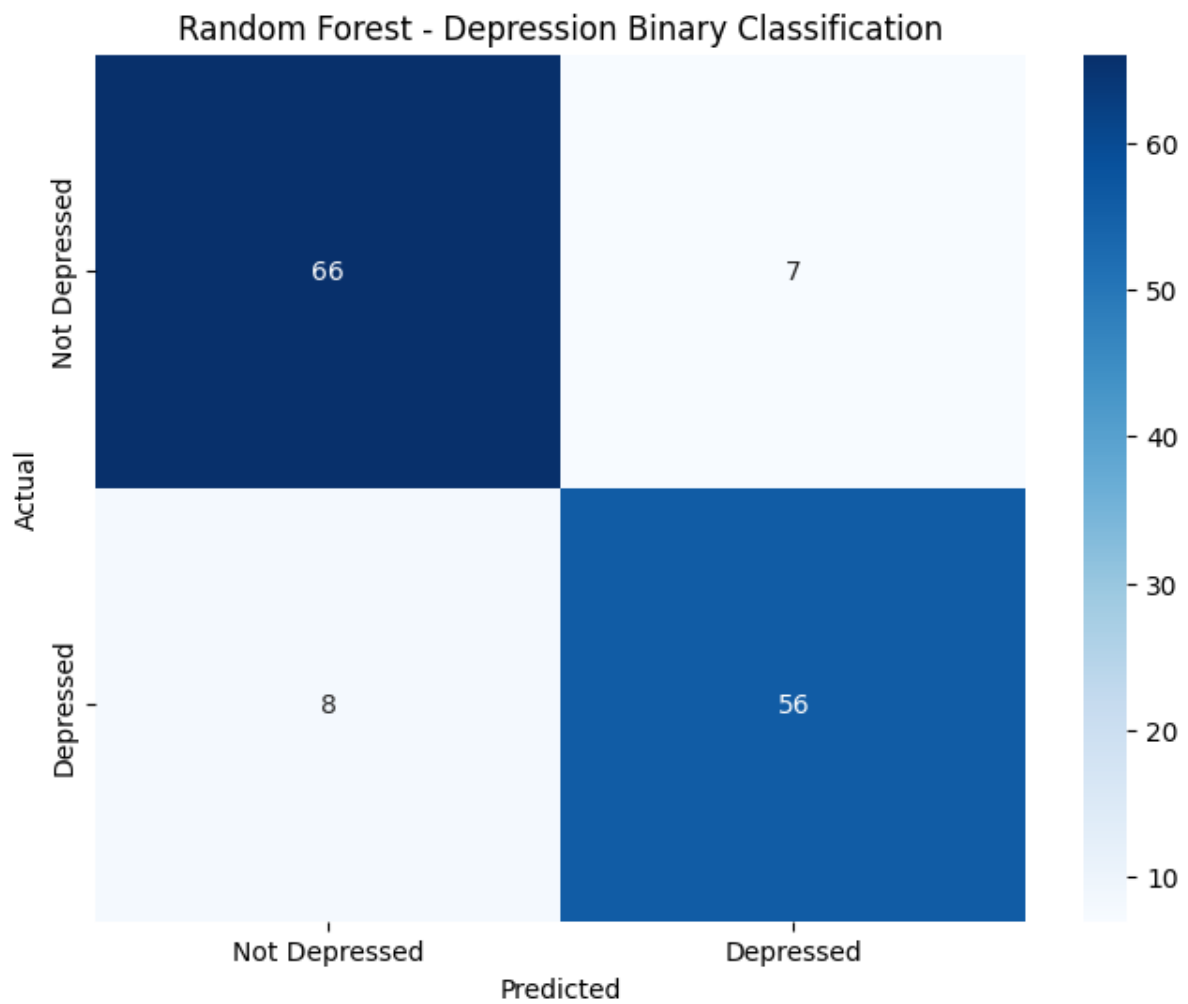


Figure 4.2: Confusion Matrix of Random Forest Classifier

Table 4.3: Classification Table of XGBoost Classifier

Class	Precession	Recall	F1-Score	Accuracy
Not Depressed	0.96	0.97	0.97	0.9635
Depressed	0.97	0.95	0.96	

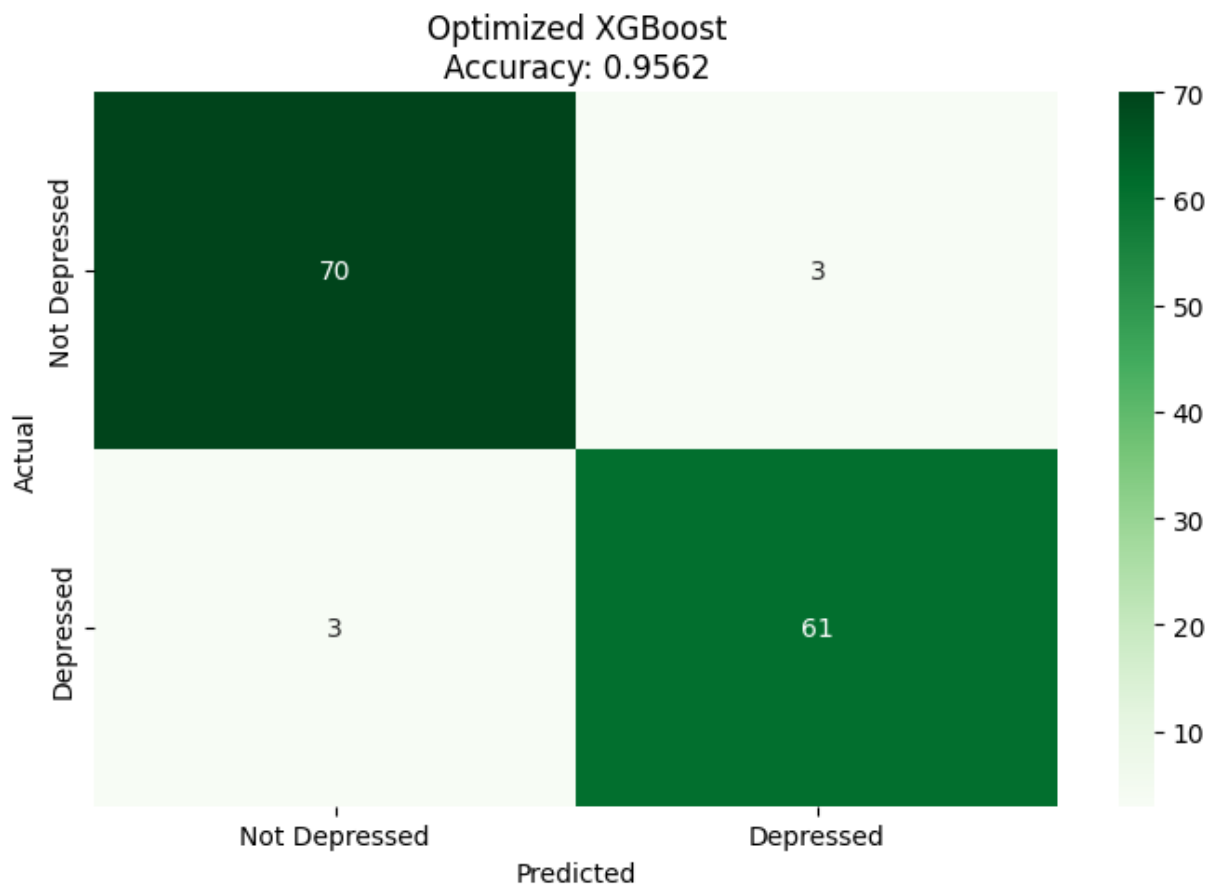


Figure 4.3: Confusion Matrix of XGBoost Classifier

Table 4.4: Classification Table of Logistic Regression

Class	Precession	Recall	F1-Score	Accuracy
Not Depressed	0.80	0.93	0.86	0.8394
Depressed	0.90	0.73	0.81	



Figure 4.4: Confusion Matrix of Logistic Regression

Table 4.5: Classification Table of Decision Tree Classifier

Class	Precession	Recall	F1-Score	Accuracy
Not Depressed	0.85	0.90	0.87	0.8613
Depressed	0.88	0.81	0.85	

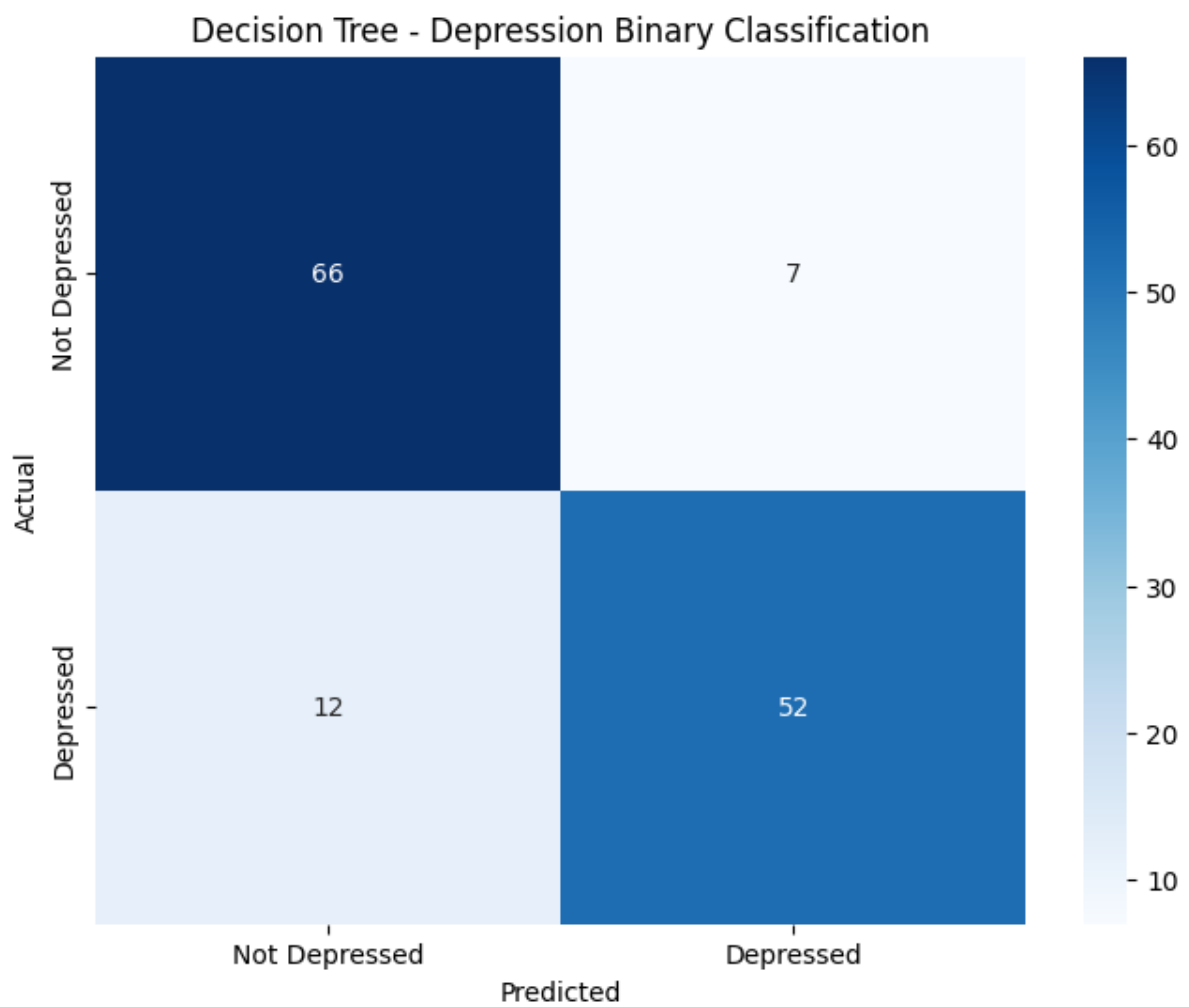


Figure 4.5: Confusion Matrix of Decision Tree

Table 4.6: Classification Table of K-Nearest Neighbors (KNN)

Class	Precession	Recall	F1-Score	Accuracy
Not Depressed	0.81	0.99	0.89	0.8686
Depressed	0.98	0.73	0.84	

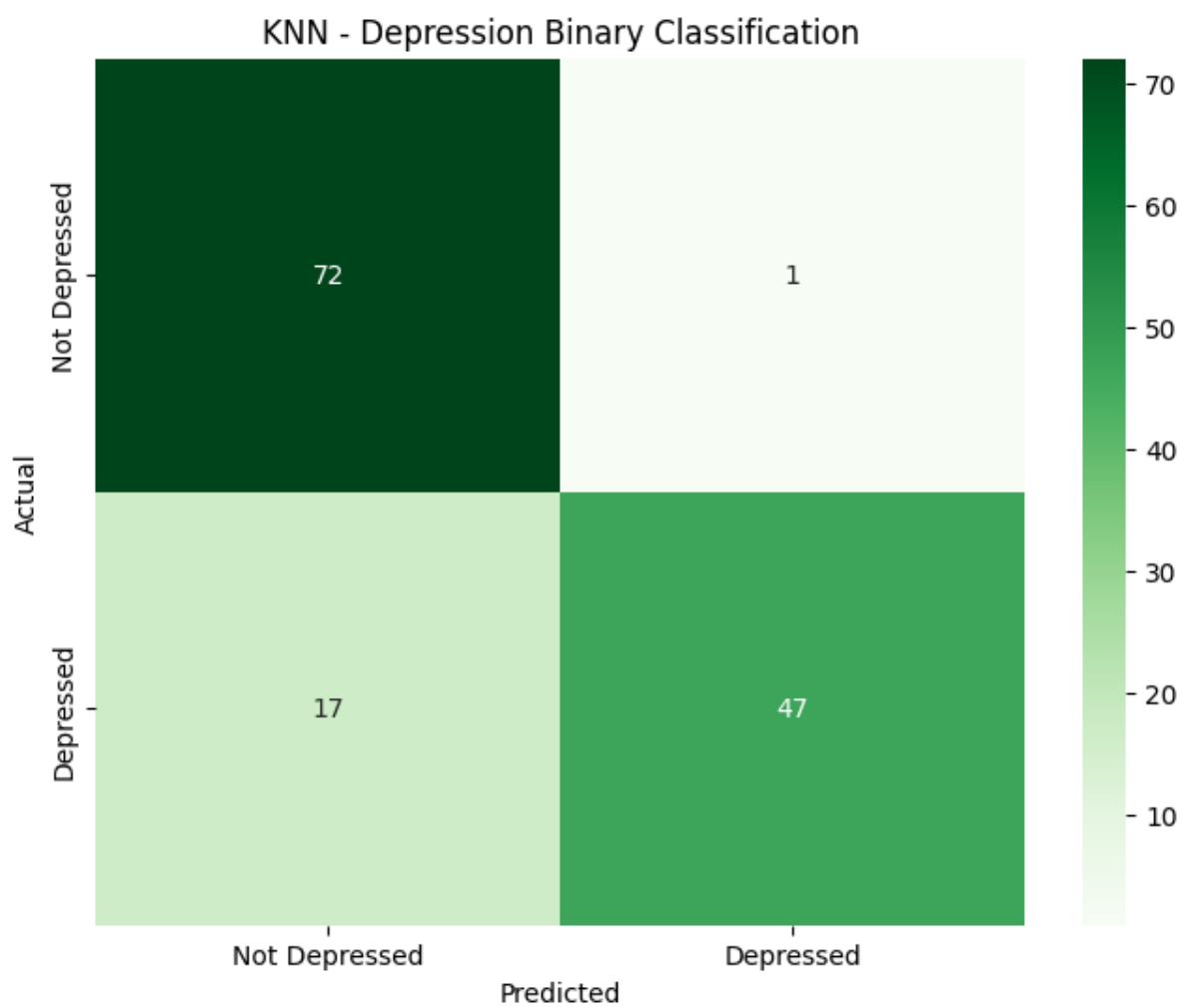


Figure 4.6: Confusion Matrix of K-Nearest Neighbors (KNN)

We tested five classic machine learning classifiers which were used to detect binary depression: Logistic Regression, Decision Tree, K-Nearest Neighbors, Random Forest, and XGBoost. Table 6.2 shows the results of comparative performance.

Table 4.7: Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
XGBoost	0.9635	0.9683	0.9531	0.9606	0.9944
Random Forest	0.8905	0.8889	0.8750	0.8819	0.9743
KNN	0.8686	0.9792	0.7344	0.8393	0.9384
Decision Tree	0.8613	0.8814	0.8125	0.8455	0.9325
Logistic Regression	0.8394	0.9038	0.7344	0.8103	0.9360

By all measures, XGBoost has been shown to perform best with an accuracy of 96.35, F1-score of 96.06 and an impressive AUC-ROC of 0.9944. The high accuracy (96.83) and recall (95.31) represent the fact that the model can be used to predict the presence of depressed individuals without wrongly giving positives.

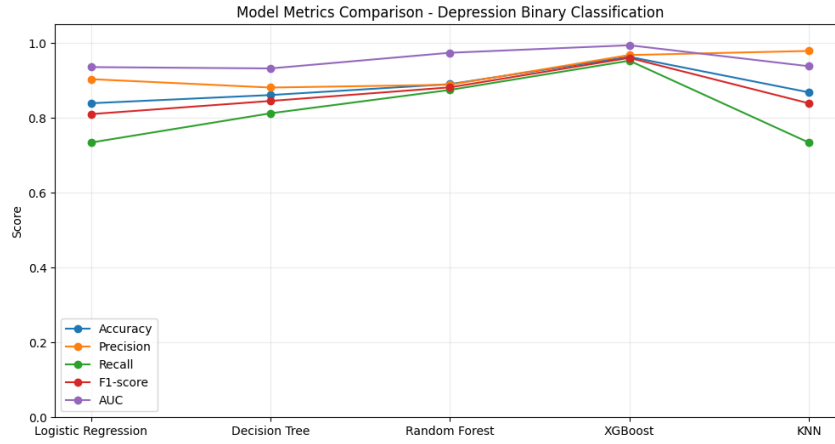


Figure 4.7: Model Metrics Comparison

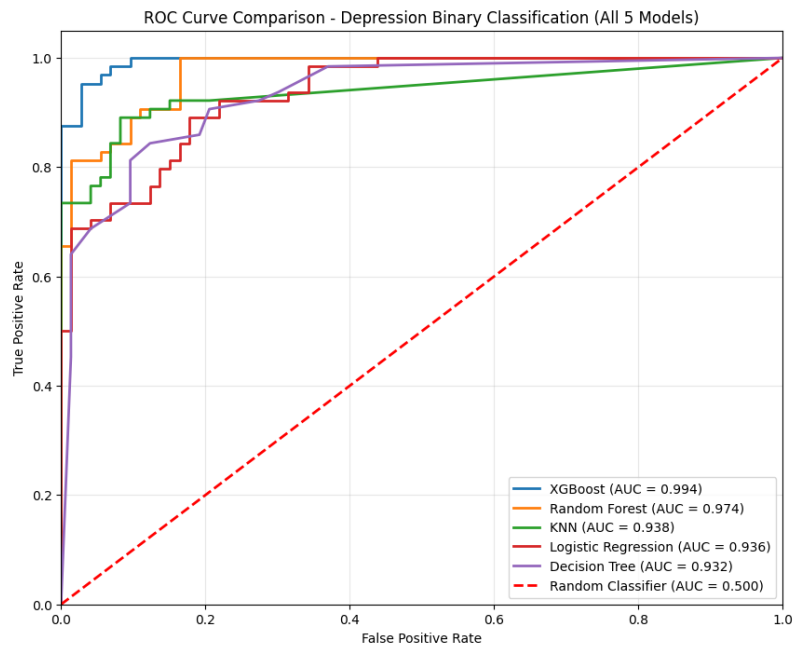


Figure 4.8: ROC Curve Comparison across all models

4.1.3 Cross Validation and Generalization Analysis

To evaluate model stability and generalization, we performed 10 cross-validation of the XGBoost classifier using stratified cross-validation. The summary of results (Table 6.3) shows that the performance was high and consistent across folds, and the average accuracy

was 94.57 per cent and the average of AUC-ROC was 0.9904. The standard deviations are relatively small in all metrics which means high variance and strong generalization.

Table 4.8: 10-Fold Cross-Validation Results (XGBoost)

Metric	Mean	Standard Deviation	95% CI
Accuracy	0.9457	± 0.0294	[0.8869, 1.0045]
AUC-ROC	0.9904	± 0.0074	[0.9756, 1.0052]
F1-Score	0.9414	± 0.0318	[0.8779, 1.0049]
Precision	0.9561	± 0.0374	[0.8814, 1.0308]
Recall	0.9286	± 0.0434	[0.8418, 1.0154]

The test accuracy (96.35) and the cross-validation mean accuracy (94.57) were separated by a small difference of 1.78 which is not significant overfitting.

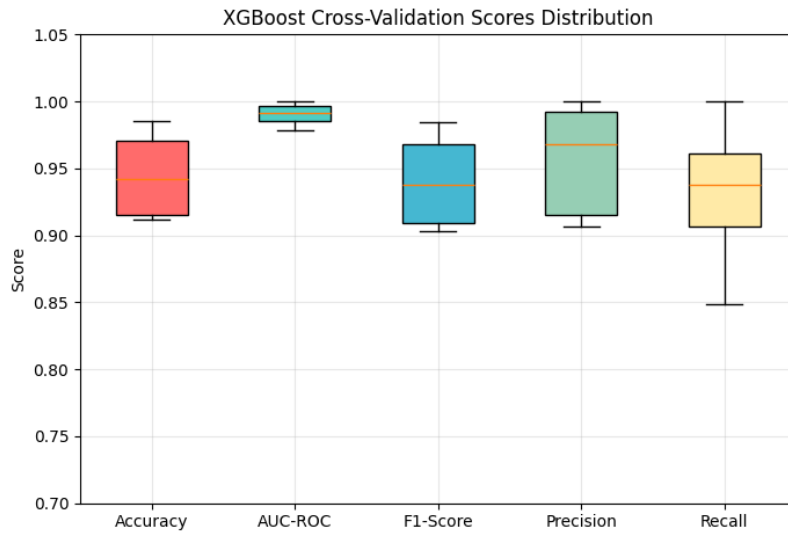


Figure 4.9: XGBoost Cross-Validation Fold Accuracy Distribution

4.1.4 Hyperparameter Optimization Effects

The crafted XGBoost configuration was obtained after hyperparameter optimization with the help of GridSearchCV that demonstrated the refined XGBoost configuration presented in Table 4.4.

Table 4.9: Refined XGBoost configuration

Parameter	Optimal Value
n_estimators	150
max_depth	7
learning_rate	0.1
subsample	0.8
colsample_bytree	0.8
min_child_weight	3

The optimized model had a minor yet significant improvement in AUC-ROC that rose between 0.9944 to 0.9957 and retained the same test accuracy. This finding supports the fact that discriminative gains can be further increased with respect to careful tuning of model depth, ensemble size, and the sampling parameters without hampering generalization. Notably, the optimization gains were in line with the cross-validation performance, which is a strong indicator that the tuned model is reliable.

4.1.5 Feature Importance and Explainability

The analysis of feature importance was performed with XGBoost importance scores, Recursive Feature Elimination, mutual information, and SHAP values. In all the strategies, the engineered feature TotalSymptomScore was the most significant predictor, which confirmed our approach of aggregating symptoms. The PHQ-9 items associated with sleep disturbance, changes in appetite, difficulty concentrating, and self-harm ideation were also among the top-ranking features of an individual, which corresponds to the known clinical studies on the symptomatology of depression.

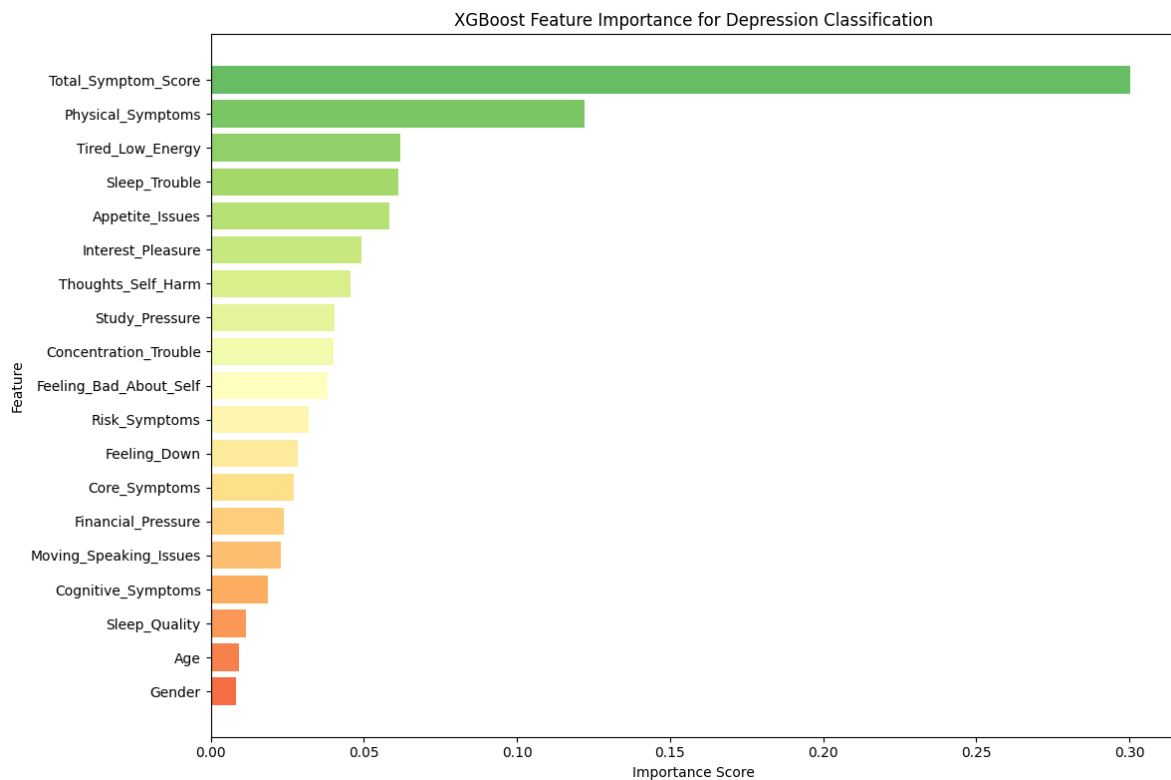


Figure 4.10: XGBoost Feature Importance for Depression Classification

Table 4.10: Feature Selection Comparison Table

Rank	XGBoost Importance	RFE Selection	Mutual Information
1	Total_Symptom_Score (0.300)	Total_Symptom_Score	Total_Symptom_Score (0.368)
2	Physical_Symptoms (0.122)	Feeling_Down	Sleep_Trouble (0.263)
3	Tired_Low_Energy (0.062)	Interest_Pleasure	Appetite_Issues (0.256)
4	Sleep_Trouble (0.061)	Tired_Low_Energy	Physical_Symptoms (0.240)
5	Appetite_Issues (0.058)	Sleep_Trouble	Tired_Low_Energy (0.236)



Figure 4.11: Feature Impact on Depression Prediction

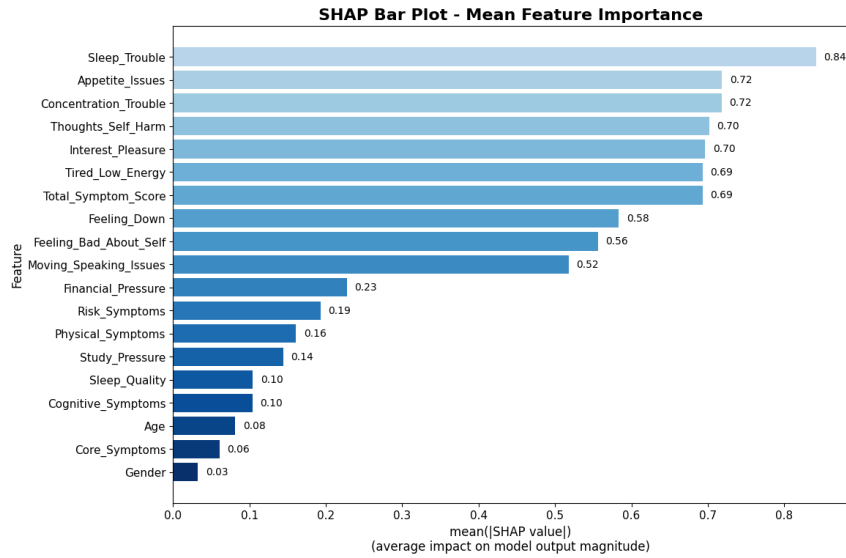


Figure 4.12: Global Feature Significance via SHAP and Cross-Validation

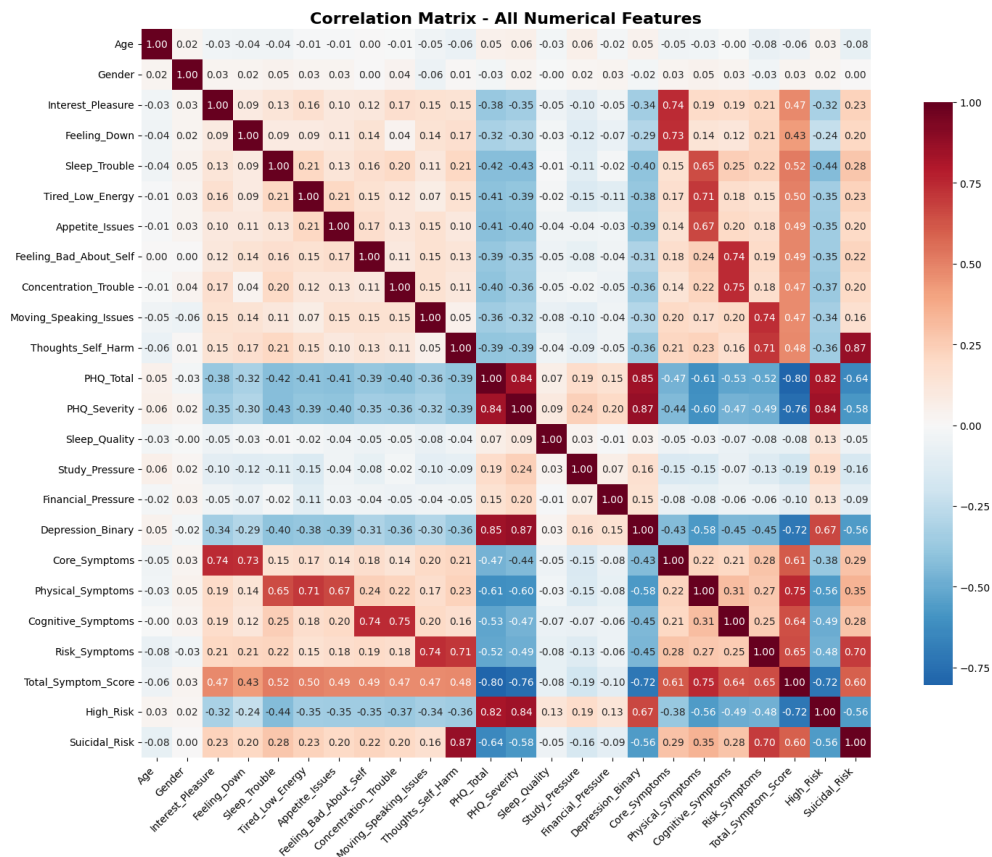


Figure 4.13: Feature Collinearity Analysis and Predictive Influence Summary

4.1.6 Probability Calibration and Clinical Risk Interpretation

Probability calibration is essential to practical clinical implementation. We compared the sigmoid (Platt scaling) and isotonic calibration.

Table 4.11: Probability Calibration Results

Calibration Method	Brier Score	AUC-ROC
Uncalibrated	0.0302	0.9957
Sigmoid (Platt)	0.0283	0.9955
Isotonic	0.0276	0.9957

This table indicates that isotonic calibration had the lowest Brier (0.0276) as it did not affect AUC-ROC, indicating it had better probability reliability.

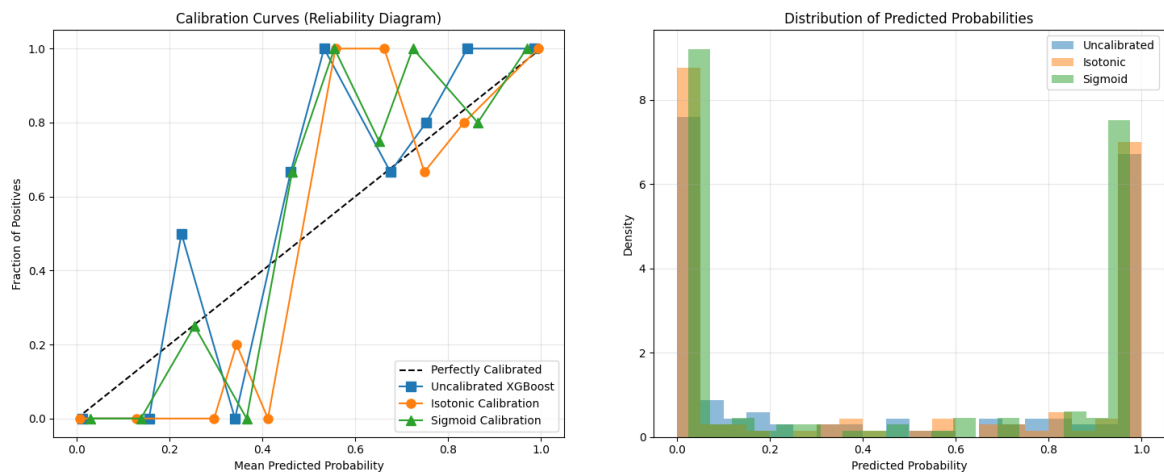


Figure 4.14: Probability Calibration Analysis and Predictive Reliability Distributions

The shift in binary classification to clinical risk assessment needs good estimates of probabilities. The analysis of the isotonic regression by reliability diagrams demonstrates that the AUC-ROC (0.9957) is the highest, whereas the Brier score (0.0276) is the lowest; this

suggests that isotonic regression is closest to the ideal calibration curve. The probability distributions give a high confidence on the model with predictions being concentrated around 0 and 1. Although the uncalibrated model already displays good discrimination, isotonic calibration additionally decreases the probability error without impacting the class separability. This enhancement is essential to clinical decision support, in which the risks that are predicted must mirror actual outcome probabilities. The isotonically calibrated XGBoost model, therefore, implies a solid basis of personalized risk stratification, where probability ranges defined in clinical terms can be used to support decisions that are amenable to action and interpretable, as opposed to black and white diagnosis.

Table 4.12: Clinical Interpretation Guide

Probability Range	Risk Level	Clinical Recommendation
0.0 - 0.2	Very Low	Routine monitoring
0.2 - 0.4	Low	Brief check-in recommended
0.4 - 0.6	Moderate	Further assessment needed
0.6 - 0.8	High	Clinical intervention recommended
0.8 - 1.0	Very High	Immediate attention required

4.1.7 Explainable AI for Clinical Transparency

To increase further trust and transparency, we used SHAP and LIME explanations in the individual-prediction level. Dependence plots like SHAP, waterfall plots, and bar charts by the LIME visualize the contribution of individual symptoms to a particular prediction. Such

explanations will enable clinicians to confirm the model reasoning is consistent with clinical judgment, which is a crucial condition of ethical AI implementation in mental health.

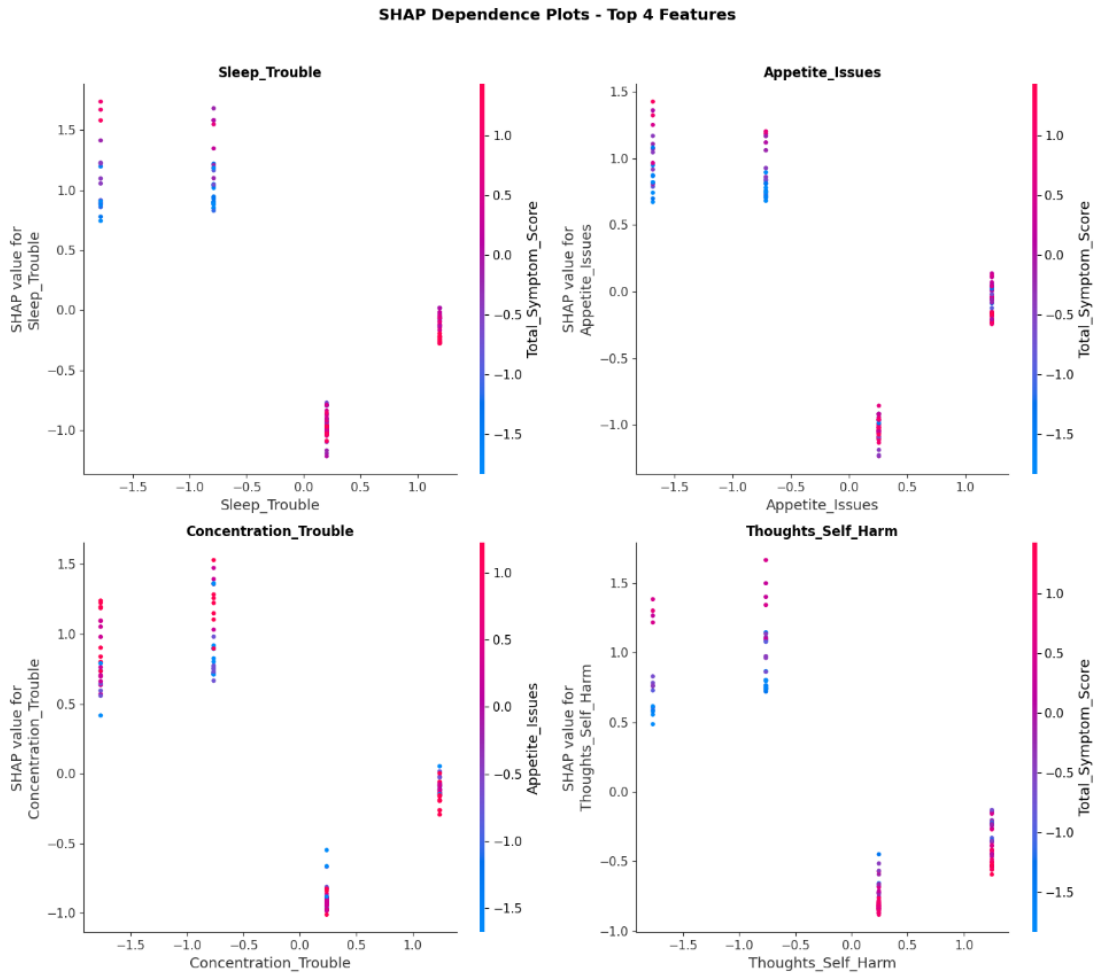


Figure 4.15: SHAP Dependence Plots - Top 4 Features

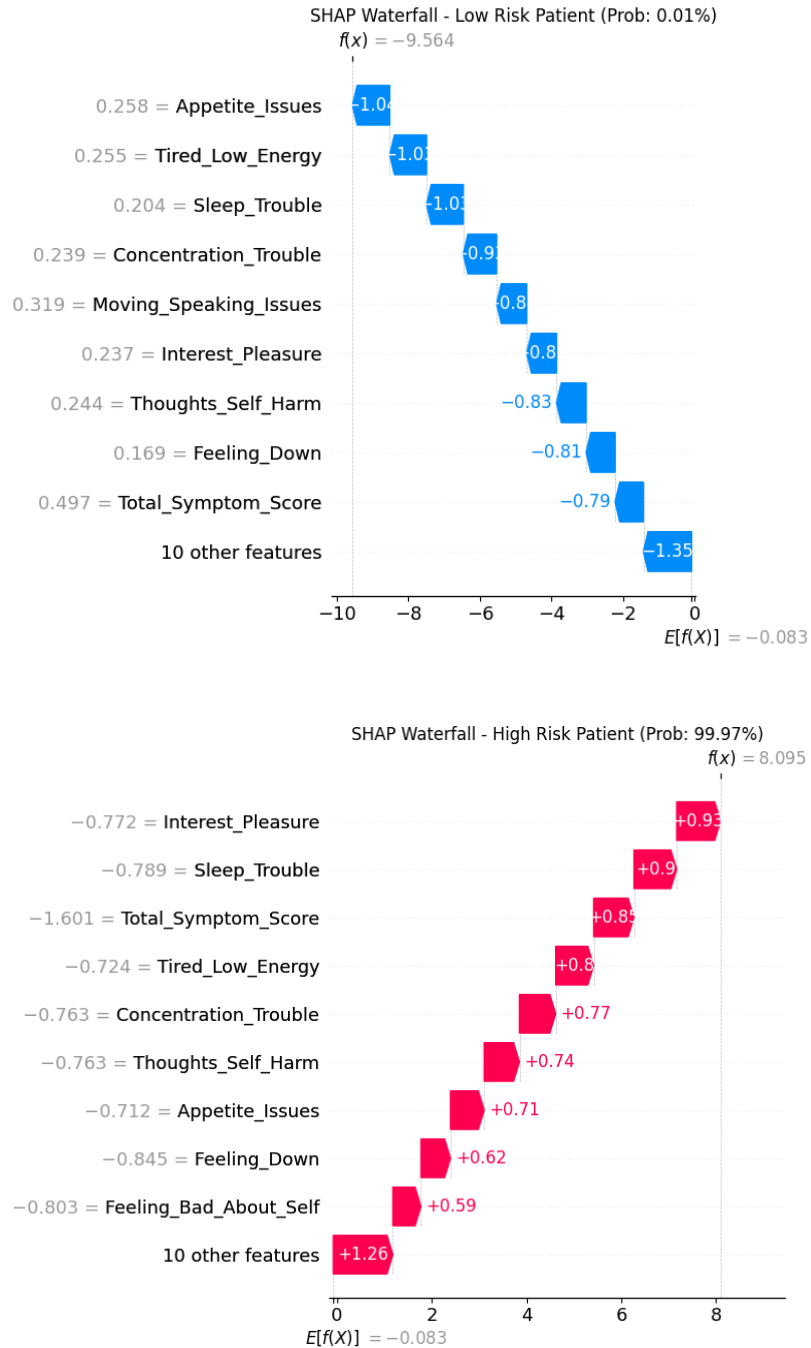


Figure 4.16: Local Interpretability via SHAP Waterfall (High vs. Low Risk)

The interpretability of the XGBoost model at the decision level is explained with SHAP waterfall plots, which break down individual predictions into cumulative feature contributions against the base value ($E[f(X)] = -0.083$). In a low-risk example, the model

yields a depression probability of almost zero (0.01%), which was caused by a strong negative indication of such a feature as AppetiteIssues and TiredLowEnergy, which means that the lack of core symptoms is a decisive exclusion aspect. Contrarily, the high-risk example produces a 99.97 predicted probability, with interestPleasure, SleepTrouble, and TotalSymptomScore taking up the dominant roles and leaving the output of the $f(x) = 8.095$. The combination of these opposite examples draws the conclusion that the model is always focused on clinically significant symptomatology, providing clear, patient-specific explanations, which are congruent with the generally accepted psychiatric norms.

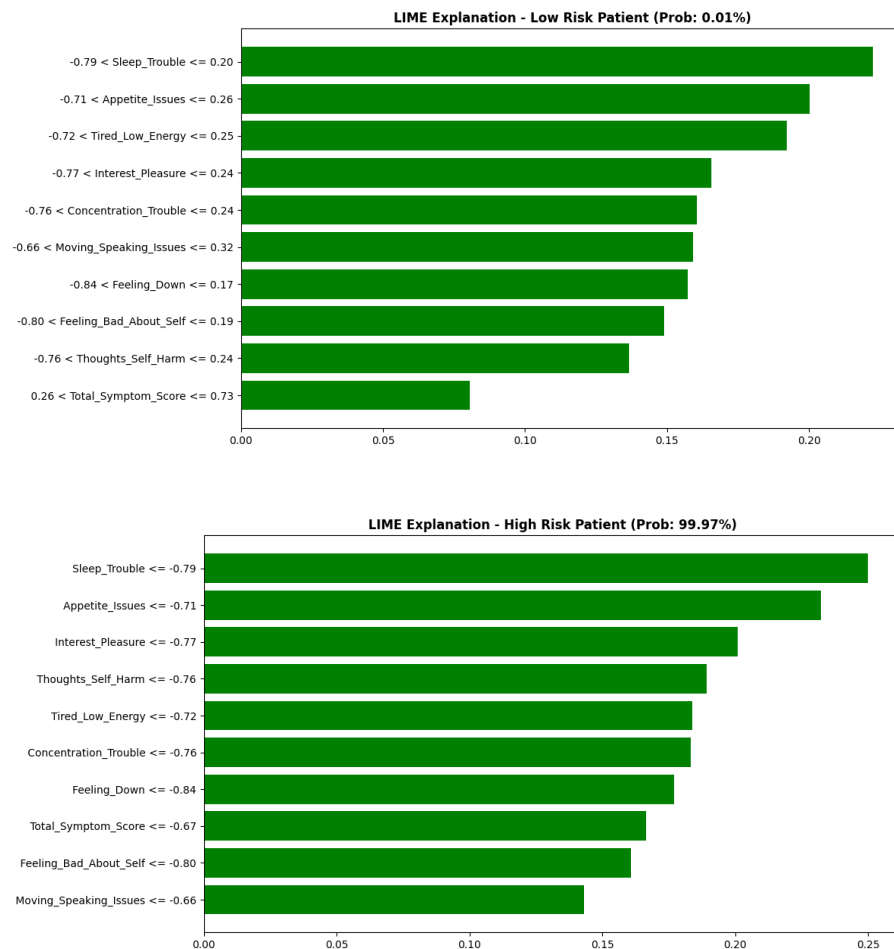


Figure 4.17: LIME Local Interpretability (Low vs. High Risk)

The model uses low- and high-risk predictions as explained in LIME through examples of two opposite cases. In the case of low risk, (0.01%), the prediction is facilitated by feature values being within non-symptomatic ranges of SleepTrouble, AppetiteIssues, TiredLowEnergy, and InterestPleasure, which play a negative role and inhibit depression risk. These restricted areas indicate clinically realistic thresholds of a non-symptomatic profile as opposed to arbitrary model behavior.

Conversely, the high-risk scenario (99.97) is characterized by extreme levels of symptoms, and significant input factors are disturbed sleep, anorexia nervosa, anhedonia, and cognitive pointers. The model has continually focused its attention on fundamental psychiatric aspects, and it results in a logical continuum between severity of symptoms and predictability.

The LIME explanations are like SHAP-based results in both profiles, and it proves that the consistency and clinically meaningful reasoning is similar regardless of interpretability tools. This consistency is useful in showing the internal validity of the model and showing that it is an open, decision-support tool to use in psychiatric assessment, as opposed to a black-box classifier.

4.2 Results of Bengali Social Media NLP Models

4.2.1 Dataset Characteristics

The Bengali NLP dataset included 2,841 short social media sentences which were annotated with PHQ-9-consistent symptoms categories and dichotomous severity ratings.

Table 4.13: Characteristics of NLP Dataset

Characteristic	Value
Total Sentences	2,841
Features	Category, Sentence, Severity Level
Classes	Depressed, Not Depressed
Max Token Length	65 tokens
Training Set	1,988 (70%)
Validation Set	426 (15%)
Test Set	427 (15%)

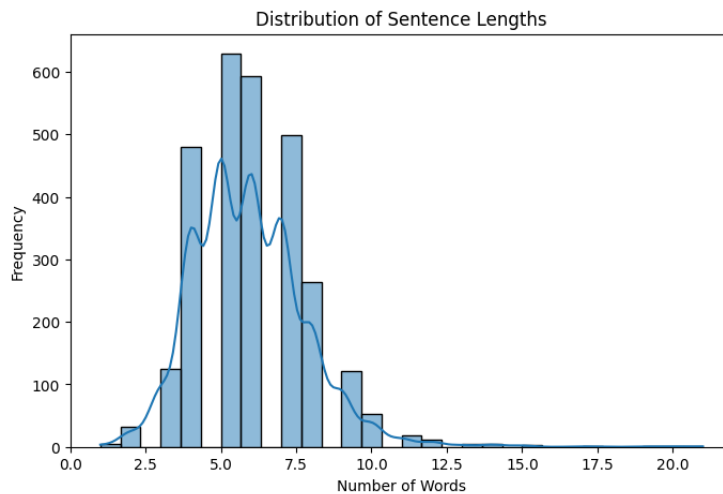


Figure 4.18: Distribution of Sentence Lengths in the Dataset



Figure 4.19: Word Cloud for the Non-Depressed Class

The visualization shows commonly used Bangla words that are related to positive emotional coloring, stability, and regular operation. The words "ভালো" (good), "সময়" (time), "ঘুম" (sleep), and "মন ভালো" (feeling well) are used sparingly which is an indication of the topics of healthy mood control, being active in everyday life, and being healthy. The prevalence of positive and neutral words reflects the presence of linguistic profile that has had an impact on people who have recorded non-clinical emotional conditions.



Figure 4.20: Word Cloud for the Depressed Class

This diagram illustrates the language patterns of the answers, which are denoted as depressing. The prevailing terms are emotional tension, diminished functioning and mental

load. The words like **মন খারাপ** (feeling down), **ঘুম সমস্যা**(sleep problems), **কাজ** (work problem), **শরীর ক্লান্ত লাগছে** (physical exhaustion) are presented with high frequency that denotes some recurring themes of fatigue, disrupted routine and emotional instability. The usage of such terms as repetitive struggle and low energy indicates the existence of a language profile that corresponds to the depressive symptomatology as the routine activities and motivation are constantly questioned.

4.2.2 Base BERT Model Performance

The multilingual BERT model was trained in ten epochs. The trends of training dynamics (summarized in Table 6.9) reflect constant loss reduction and level validation accuracy (79%).

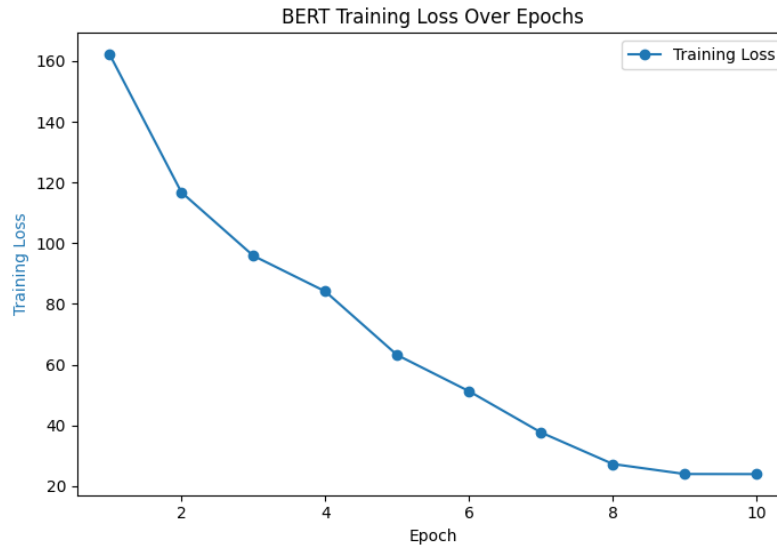


Figure 4.21: Base BERT Model Training Loss

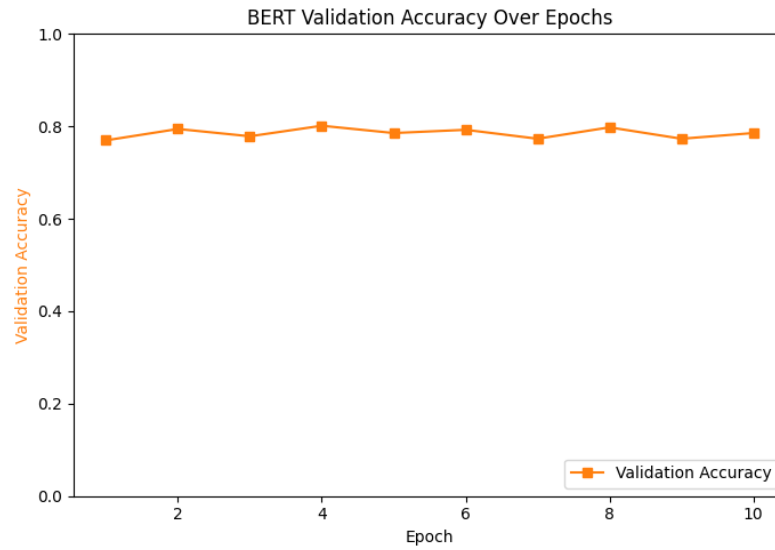


Figure 4.22 Base BERT Model Validation Accuracy

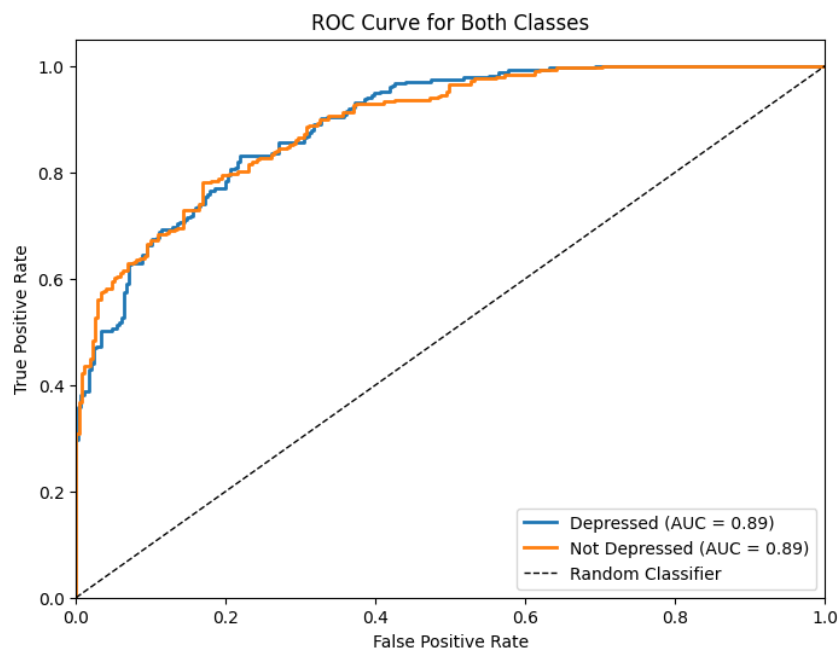


Figure 4.23 ROC Curve for Base BERT

Table 4.14: Base BERT Classification Report

Class	Precision	Recall	F1-Score	Accuracy
Depressed	0.74	0.85	0.79	0.79
Not Depressed	0.84	0.73	0.78	

4.2.3 Stacked Ensemble Model Architecture

The ensemble method applies to a wide range of architectures to bring out various textual variations. The component models are broken down as below and what each of them contributes to the eventual prediction.

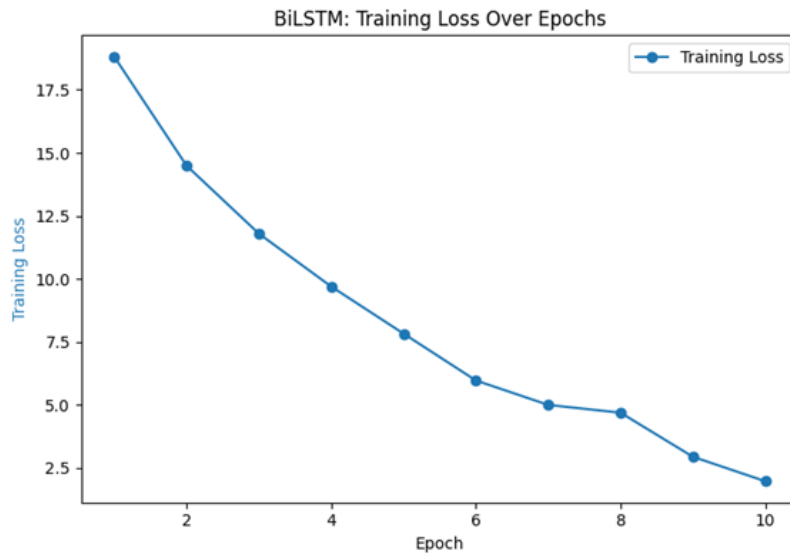


Figure 4.24: BiLSTM Training Loss

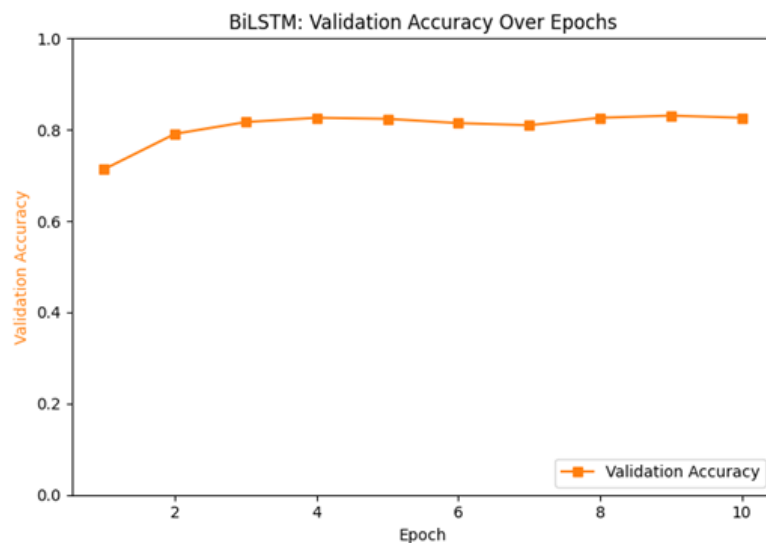


Figure 4.25: BiLSTM Validation Accuracy

4.2.4 Ensemble Learning for Performance Enhancement

We overcame these shortcomings by building a stacked stack of BERT, BiLSTM and Random Forest model, where the meta-learner is Logistic Regression.

Table 4.15: Base Models vs Ensemble Performance

Model	Accuracy	Precision (D)	Recall (D)	F1-Score (D)	Precision (ND)	Recall (ND)	F1-Score (ND)
BERT	0.9578	0.94	0.97	0.96	0.97	0.94	0.96
LSTM	0.8103	0.80	0.82	0.81	0.82	0.80	0.81
Random Forest	0.7728	0.76	0.80	0.78	0.79	0.75	0.77
Ensemble	0.9578	0.95	0.96	0.96	0.96	0.95	0.96

D = Depressed, ND = Not Depressed

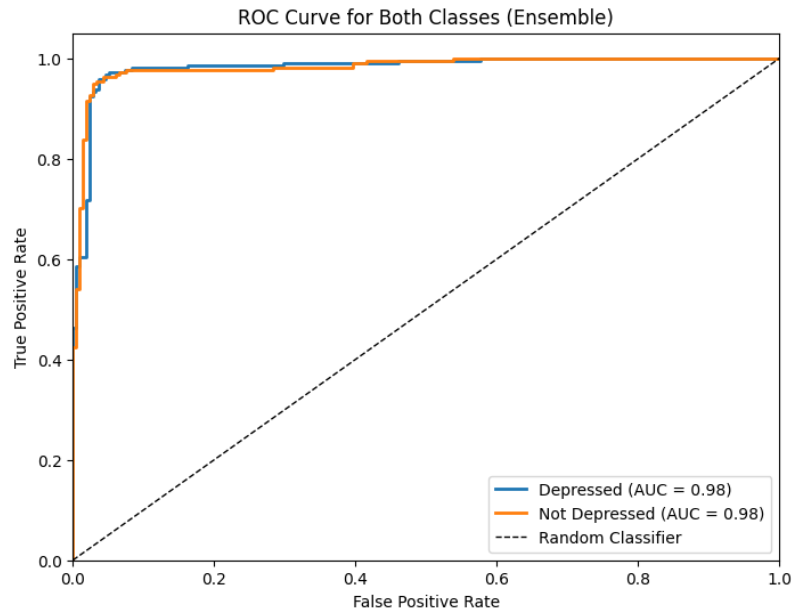


Figure 4.26: ROC Curve of Ensemble Model

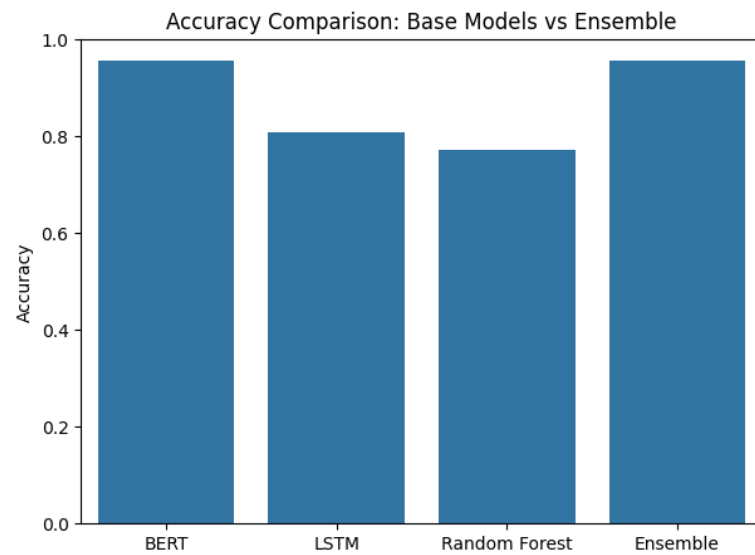


Figure 4.26 Accuracy Comparison Between Base Models and Ensemble Approach

This figure compares the predictive results of the single baseline models and the ensemble method suggested. Out of the standalone architectures, BERT is the most accurate, with LSTM and Random Forest being slightly less. The ensemble model is superior to all the individual models, which means that the aggregated decision-making gives a stronger

estimate of depressive status. The latter enhancement underscores the advantage of using heterogeneous learning signals to minimize model-specific bias and stabilize overall reliability in prediction.

4.3 Comparative of PHQ-9 Based and Bengali NLP-Based Approaches

To achieve the goal of comparing structured and unstructured depression detection paradigms, we used the best-performing XGBoost PHQ-9-based model and the best Bengali NLP ensemble. PHQ-9 model had a little better accuracy (96.35) and AUC-ROC (0.9944) when compared to the NLP approach (95.78% accuracy; AUC-ROC approximate 0.98) due to the clinically validated and highly discriminative items of PHQ-9.

The Bengali NLP ensemble can also perform similarly with unstructured text, albeit with a slightly higher number of hits in the depressed group (96% vs. 95.31%), which suggests that it is highly sensitive to depression manifested through language. Precision and F1-scores do not differ between the two models, and this point shows that more sophisticated NLP approaches were effective in identifying clinically significant signals.

The PHQ-9 model is less expensive to compute methodologically, can be trained and inferred in a shorter time, and involves less interpretability through SHAP and LIME, which makes it suitable to deploy in a clinical setting. Conversely, the NLP ensemble with BERT, LSTM and Random Forest involve GPU usage, has a high latency, and is specific to a language and thus needs a close cultural verification.

Lastly, there is a difference between data acquisition in the two approaches. PHQ-9 is an active and questionnaire-based system, and NLP allows passive monitoring of text on social media of interest, which have significant implications in scalability, user burden, and longitudinal mental health surveillance.

4.4 Clinical Implication

4.4.1 PHQ-9 Clinical Deployment

The outstanding performance statistics of the XGBoost model (96.35% accuracy, 0.9944 AUC-ROC, Brier Score 0.0276) fulfill the high demands of the clinical decision support systems. The balanced precision (96.83) and recall (95.31) guarantee that there would be few false negatives (missed cases of depression), and false positives are regulated.

Table 4.16: Risk Stratification Framework

Risk Tier	Probability	Clinical Action	Follow-up
Tier 0	0.0-0.2	Routine Monitoring	Annual Screening
Tier 1	0.2-0.4	Brief check-in	3–6-month follow-up
Tier 2	0.4-0.6	Comprehensive Assessment	1–3-month follow-up
Tier 3	0.6-0.8	Clinical Intervention	Weekly monitoring
Tier 4	0.8-1.0	Immediate Attention	Crisis Protocol

We combine SHAP and LIME explainability models to give transparent predictions at an instance-level of our model. These approaches enable us to measure the value of features to each prediction, thus enabling clinicians to have an idea about not only the predicted value but also the reasoning behind the choice taken by the model. The system assists with the clinical intervention targeted and enables the planning of the treatment individually by clearly defining the risk factors related to a symptom. Additionally, the fact that interpretable explanations are available is consistent with regulatory requirements of medical artificial

intelligence systems, such as requirements related to the U.S. Food and Drug Administration (FDA) approval processes and CE marking of medical devices under European medical device regulations.

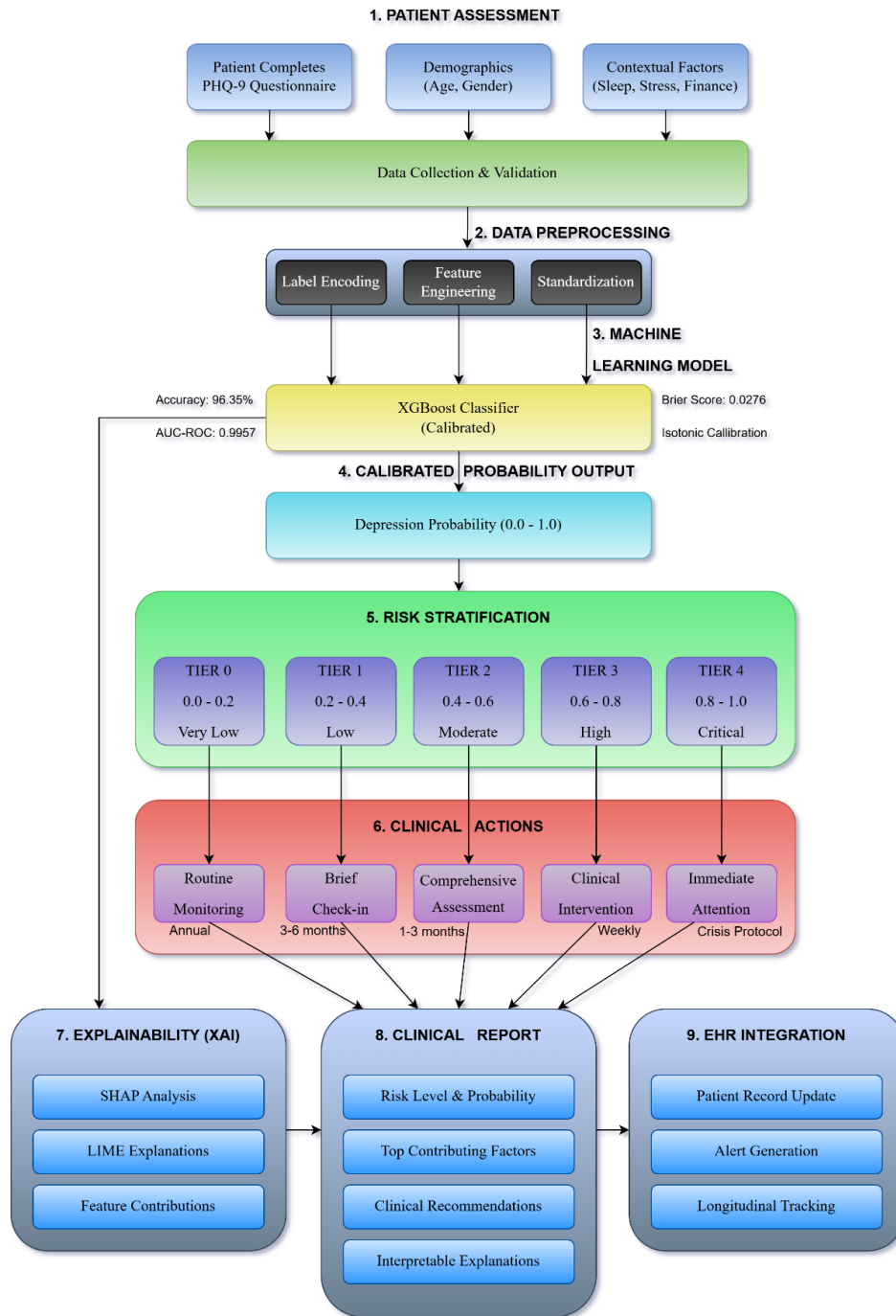


Figure 4.27 Clinical Decision Support Workflow for Model Integration

4.4.2 Bengali NLP Model Application

The good classification performance of Bengali NLP ensemble model (95.78%) facilitates its potential application in mental health surveillance conducted in large scale through social media. Using linguistic patterns and context-based cues introduced in publicly accessible text-based data, the model allows tracking population-wide depressive speech without the involvement of individuals. The given capacity facilitates the identification of the at-risk groups early via identification of language indicators as the signs of emotional distress, negative self-referential language, and cognitive symptoms of depression. Moreover, the model allows conducting temporal analysis of mental health patterns, which allows researchers and authorities responsible for population health to monitor changes in psychological well-being at the community level and over time. The technique can also be used to geographically map mental health risk when it is combined with location metadata on an aggregate level, which can subsequently be used to inform public health interventions in a region.

4.4.3 Early Warning System

In addition to retrospective surveillance, the Bengali NLP model can be implemented as an early warning system in detecting depression as a real-time system. Through continuous streamline analysis of social media, the system will be able to detect posts most likely to contain depressive messages and send threshold-based awareness. This type of alerts can be used as a signal to downstream intervention strategies, such as mental health helplines, crisis response measures, or community support services. Notably, the specified functionality can be applied in a privacy-resistant way as it can be applied to the anonymized or aggregated data without profiling individuals. By so doing, the system will provide the balance between

mental health assistance when it is necessary and ethical aspects concerning the privacy of data and responsible AI use.

4.4.4 Community Mental Health Screening

The suggested NLP-based model can also provide an affordable and scalable alternative to mental health screening at the community level. As opposed to conventional questionnaire-based tests, where participants must deal with the test and results directly, the NLP method allows indirect screening based on the data of the naturally occurring language. This feature comes in handy especially in helping to screen underserved or marginalized groups that might lack access to clinical screening aids and are on online platforms. The model can be used to direct the strategic distribution of mental health resources and to target outreach efforts by highlighting communities at risk of depression. Also, the longitudinal analysis of the language patterns prior to and after the intervention gives a possibility to evaluate the effectiveness of the interventions over the period, which offers the data-driven foundation of the evaluation and further development of mental health programs.

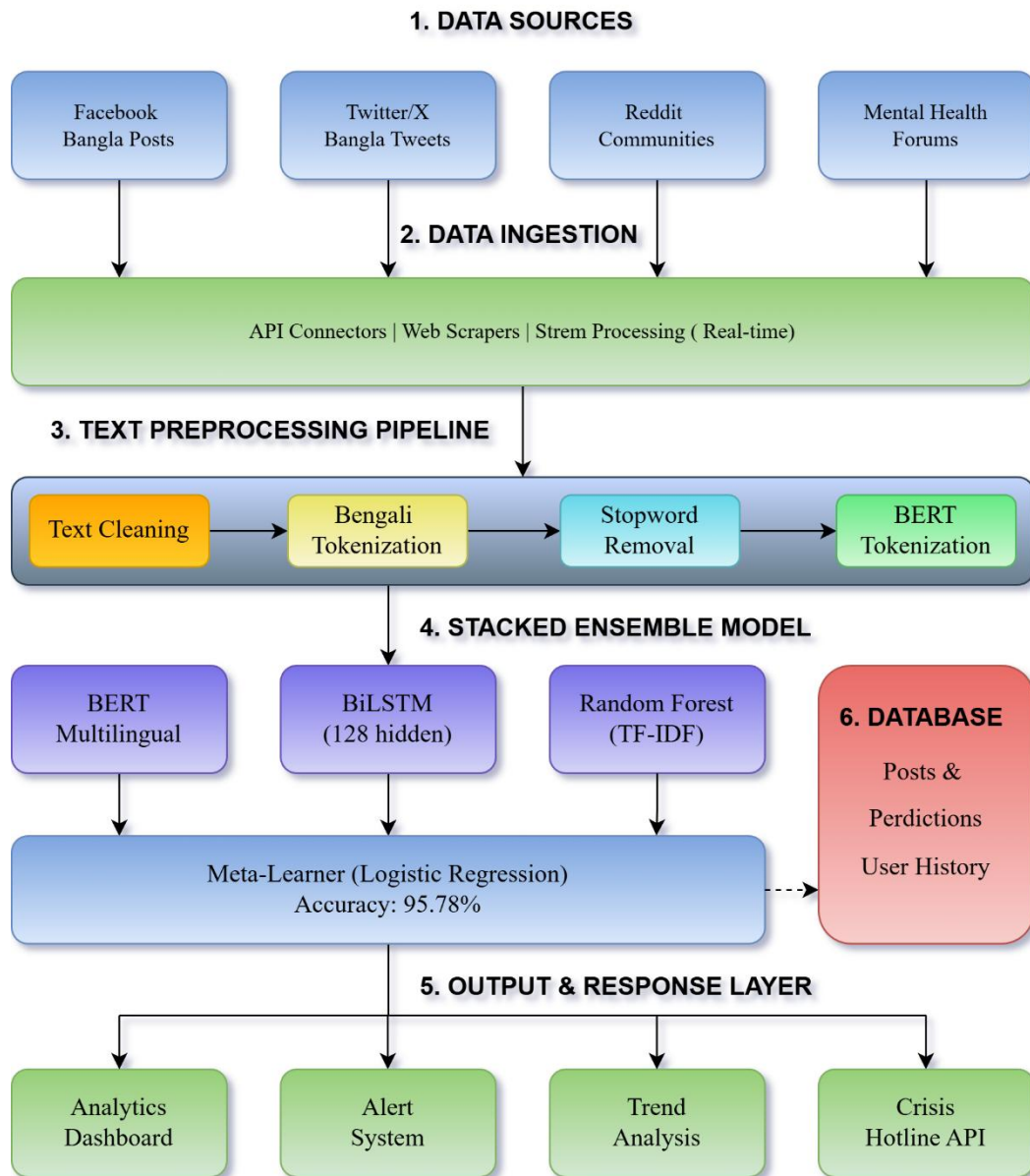


Figure 4.28 System Architecture for Social Media Mental Health Monitoring

4.5 Integrated Dual-Modality Approach

We suggest a dual-modal model that is integrated to combination of structured PHQ-9 based machine learning and unstructured Bengali NLP based analysis to facilitate full and persistent depression screening. These modalities are rather complementary to each other, as

they fill in the temporal, behavioral, and contextual gaps that are caused by single-source assessment methods.

4.5.1 Complementary Strengths

The PHQ-9 model gives the clinician-directed, standardized tests at specific time slots whereas the NLP model allows passive continuous observations of the linguistic expressions on social media. Screening based on questionnaires can be influenced by response bias and low frequency of assessment and analysis based on language takes into consideration spontaneous emotional disclosure with low reporting bias. These methods when combined provide greater coverage, effective use of resources, and cross-validation of risk indicators as summed up in Table 4.12.

Table 4.17: Complementary Strengths of the Dual-Modality Framework

Aspect	PHQ-9 Model	NLP Model	Combined Value
Assessment Type	Active	Passive	Comprehensive coverage
Frequency	Periodic	Continuous	Gap filling
Data Source	Clinical	Social media	Multiple touchpoints
Response Bias	Present	Minimal	Cross-validation
Resource Need	Moderate	Low (automated)	Efficient screening

4.5.2 Integrated Clinical Workflow and Outcomes

The Bengali NLP model, in the proposed workflow, is used to monitor the social media content continuously and highlight the signals of the high probability of depression to be

followed up on. The flagged people are further subjected to clinical analysis based on PHQ-9, and the XGBoost model offers decision support. Passive NLP surveillance is done at the time of clinical visits and provides the opportunity to detect the development of symptoms or relapse at the earliest. This combined strategy should assist in earlier intervention, time to treatment, better longitudinal monitoring and better cost-effective management of mental health at population level.

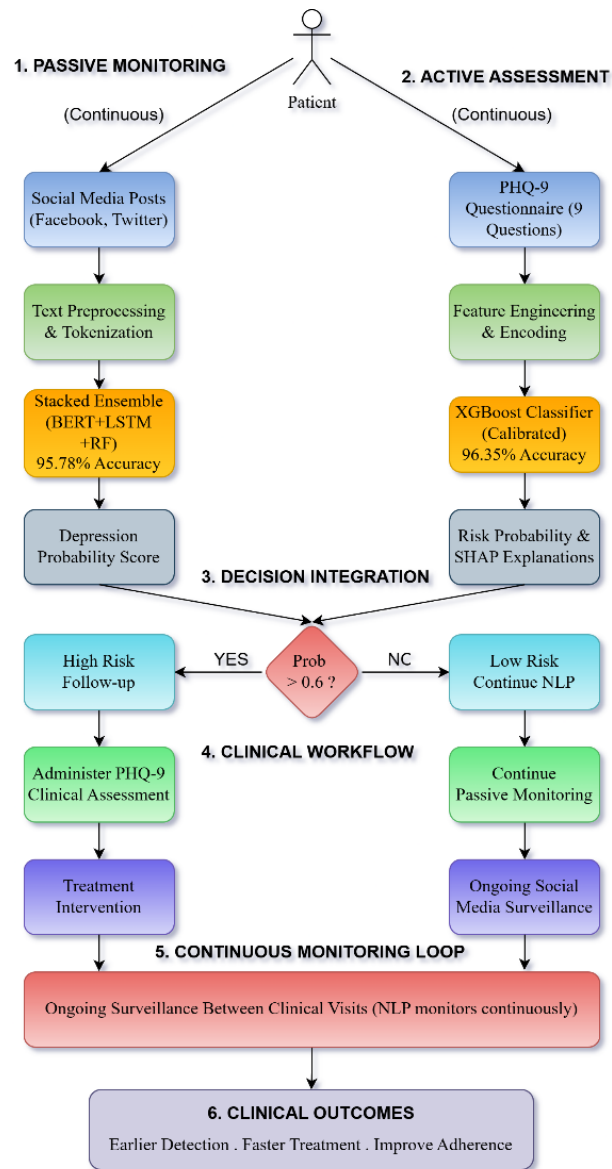


Figure 4.29 Integrated Clinical Workflow for Dual-Modality Depression Screening

4.6 Comparison with Previous Studies

The following table shows the main methodological and performance distinctions between the two pipelines that were utilized in this study: PHQ-9 based traditional machine learning and Bengali social media-based NLP modeling.

Table 4.18: Summary of Our Approach

Aspect	PHQ-9 Based ML	Bengali social media NLP
Dataset	682 Bangladeshi university students	2,841 manually curated Bengali sentences
Models Used	Logistic Regression, DT, KNN, RF, XGBoost	RF (TF-IDF), LSTM, mBERT, Stacked Ensemble
Best Model	XGBoost	Stacked Ensemble (mBERT + LSTM + RF)
Accuracy	96.35%	95.78%
AUC-ROC	0.9944	High (ROC presented for BERT & ensemble)
Cross-Validation	10-fold stratified	80/20 split
Calibration	Isotonic + Platt Scaling	Not emphasized
Explainability	SHAP + LIME	Limited (mainly performance-based)
Novelty	Dual-modality comparative framework	Bengali low-resource transformer modeling

The table below is a comparison of our work with related works on key related studies.

Table 4.19: Comparative Study with Previous Works

Study	Data Source	Methodology	Key Results	Comparison with Our Study
Haq et al. (2022)	Bangladeshi university students (PHQ-9 & GAD-7)	Traditional ML models (Random Forest, SVM)	Random Forest achieved the highest accuracy for depression prediction	This research applies to the same setting of population. Our methodology is better than that of our study because we introduce XGBoost, stratified cross-validation, probability calibration, and SHAP/LIME-generated interpretability.
Glavin et al. (2022, 2023)	PHQ-9 datasets	Feature optimization and item selection using ML	Optimized PHQ-9 item combinations improved screening efficiency	They base their work on the choice of fewer PHQ-9 items. Compared to that, our study extends the modeling by applying full-feature training and engineered symptom grouping and risk indicators.
Alves et al. (2025)	PHQ-9 + clinical notes (EMR)	Ensemble ML combining structured and unstructured data	High AUC values and scalable automated screening	Here our activity is the only one that conducts a direct comparative study of structured PHQ-9 ML and Bengali social media NLP in one unified framework which makes our approach more robust.

Colledani et al. (2024)	PHQ-9 + demographics	Decision Tree models for personalized assessment	Clinically interpretable decision paths	They are concerned with decision tree interpretability. Our results in XGBoost (96.35% accuracy) perform better on our study but we try to explain what happens using SHAP and LIME.
Hossain et al. (2022)	Bengali Facebook posts	LIWC + Random Forest	Identified depressive linguistic markers successfully	Their effort validates pointers of linguistic depression in Bengali. We promote it with LSTM and mBERT, and the stacked ensemble technique, and obtain 95.78% accuracy.
Skaik (2021)	Social media posts	Linguistic feature analysis	Population-level depression prediction was feasible	The main purpose of this work is to show the possibility of it. Our work offers a conscientious comparative analysis through use of transformer-based contextual modeling, as well as a more vigorous application of experimental validation to provide solid and dependable outcomes.
Razavi et al. (2020)	Smartphone and social metadata	ML classification models	Passive monitoring for depression detection was feasible	Their work is concerned with passive behavioral metadata. Our research is concerned with linguistic content and directly incorporates alignment with PHQ-9 based clinical screening.

Kim & Lee (2021)	PHQ-9 validated digital datasets	ML and BERT-based approaches	High pooled AUC values across studies	This paper summarizes the good performance of ML studies. We provide Bengali low-resource language modeling and a dual-feet comparative design.
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Chapter 5. Conclusion

5.1 Conclusion

This study was a complete comparative study of automated depression detection through structured PHQ-9 questionnaire data and unstructured Bengali social media text. The study compared the relative effectiveness, interpretability, and applicability of the traditional machine learning, deep learning, and ensemble-based models in both modalities by a systematic evaluation of their effectiveness.

The results indicate that machine learning models that are trained using PHQ-9 data have a highly accurate prediction. XGBoost was probably the best-performing model among the assessed ones, as it demonstrated high accuracy, generalization properties, and sound philosophy estimations of probabilities. The additional explainability analyses of SHAP and LIME showed that the model predictions were based on clinically relevant depressive symptoms, which validates the model transparency and its role as a clinical decision-support tool.

Simultaneously, the analysis revealed that the Bengali social media text has a potential to be successfully employed in depression detection by using NLP techniques. Transformer-based models and ensemble models had managed to learn linguistic expressions of depressive symptoms, and they could perform similarly to structured questionnaire-based models. The ensemble approach allowed us to achieve robustness by integrating contextual,

sequential and statistical representations which showed the efficacy of heterogeneous modeling in low resource language conditions.

The comparative analysis showed that the PHQ-9-based models are better interpreted and standardized, which makes them very appropriate to be implemented in clinical setting, whereas the NLP-based models provide passive, continuous scaling of monitoring, and no active user intervention is required. These synergistic advantages inspired the suggested dual-modality framework that combines active clinical screening and passive linguistic monitoring to facilitate intensive and repeated assessing of depression. Altogether, the findings suggest that both structured and unstructured interventions are not adequate enough to discuss the complexity of depression in all its manifestations, but their combination provides a scalable, balanced, and ethical solution to its early detection and monitoring.

5.2 Limitations

Although this study has a strong empirical performance and systematic design, several limitations must be mentioned. To begin with, the PHQ-9-based analysis was based exclusively on the data gathered among the students at Bangladeshi universities. Although this population is very relevant based on academic and psychosocial stressors, the narrowness of demography restricts the generalization of the results to non-student, older, and clinically diagnosed populations.

Second, the structured and unstructured elements of the study rely on self-reported or manual data annotated. The responses of PHQ-9 are prone to recall bias and social

desirability effects and the manual alignment of severity levels in the Bengali NLP dataset could create inconsistencies in annotation to influence model learning.

Third, depression identification was made a binary classification problem to make sure that methodological consistency is maintained across modalities. Even though it is appropriate to use this method to screen, it fails to reflect the entire range of the severity of depression and underrepresents the borderline or subclinical cases.

Moreover, the modeling method was cross-sectional, and it was impossible to analyze time dynamics of symptoms, remission, or relapse. The NLP models were also availed to short textual social media in the Bengali language, which narrows its use to long texts, other platforms, and other languages. Lastly, whereas PHQ-9-based models are computationally lightweight, transformer-based models of NLP need to be accelerated with a GPU, thus limiting their implementation to resource-constrained settings.

5.3 Future Work

This work can be further extended in future research in several significant ways. External validity would be enhanced by validation on larger and more diverse and clinically diagnosed populations, which would facilitate broader clinical usage. The reduction of binary classification to multiclass or ordinal modeling according to PHQ-9 severity levels would allow them to stratify risks and be interpreted more effectively by the clinician.

The addition of longitudinal data, e.g. repeated PHQ-9 measurements and time-stamped social media usage would enable time modeling of symptom patterns, early relapse, and

analysis of the response to treatment. Clinical credibility would also be enhanced by use of ground truth labels as clinician verified diagnoses, structured psychiatric interviews and treatment outcomes.

Also, the suggested dual-modality framework may be extended into the multimodal system that incorporates behavioral, physiological or smartphone-based signal, as well as questionnaire and language data. Fairness, demographic bias and ethical considerations should also be expressly evaluated in future studies so that deployment can be responsible. Lastly, pilot deployments and clinical assessments will be required in the future to determine the usability, trust of clinicians, acceptance of patients, and effectiveness of the proposed system in practice.

References

1. Alves, P., Marci, C. D., Cohen-Stavi, C. J., Whelan, K. M., & Boussios, C. (2025). A machine learning model using clinical notes to estimate PHQ-9 symptom severity scores in depressed patients. *Journal of Affective Disorders*, 376, 216–224. <https://doi.org/10.1016/j.jad.2025.01.152>
2. S Vidya, Swamy, N., & H Lalitha. (2024). Leveraging Machine Learning Algorithms for Depression Prediction in College Students Using PHQ-9 Scores. 1–2. <https://doi.org/10.1109/phdedits64207.2024.10838637>
3. Kim, S., & Lee, K. (2021). Screening for Depression in Mobile Devices Using Patient Health Questionnaire-9 (PHQ-9) Data: A Diagnostic Meta-Analysis via Machine Learning Methods. *Neuropsychiatric Disease and Treatment*, Volume 17, 3415–3430. <https://doi.org/10.2147/ndt.s339412>
4. Lee, K., Lee, H.-K., Kim, S., & Kim, S. H. (2021). The relationship between circadian typology and lifetime experiences of hypomanic symptoms. *Psychiatry Research*, 298, 113788. <https://doi.org/10.1016/j.psychres.2021.113788>
5. Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2001). The PHQ-9: Validity of a Brief Depression Severity Measure. *Journal of General Internal Medicine*, 16(9), 606–613. <https://doi.org/10.1046/j.1525-1497.2001.016009606.x>

6. Sheehan, D. V., Lecrubier, Y., Sheehan, K. H., Amorim, P., Janavs, J., Weiller, E., Hergueta, T., Baker, R., & Dunbar, G. C. (1998). The Mini-International Neuropsychiatric Interview (M.I.N.I.): the development and validation of a structured diagnostic psychiatric interview for DSM-IV and ICD-10. *The Journal of Clinical Psychiatry*, 59 Suppl 20, 22–33;quiz 34-57. <https://pubmed.ncbi.nlm.nih.gov/9881538/>

7. Seal, A., Bajpai, R., Karnati, M., Agnihotri, J., Yazidi, A., Herrera-Viedma, E., & Krejcar, O. (2022). Benchmarks for machine learning in depression discrimination using electroencephalography signals. *Applied Intelligence*. <https://doi.org/10.1007/s10489-022-04159-y>

8. Mukku, L., & Thomas, J. (2023). A machine learning model to predict suicidal tendencies in students. *Asian Journal of Psychiatry*, 79, 103363. <https://doi.org/10.1016/j.aip.2022.103363>

9. Daiana Colledani, Robusto, E., & Anselmi, P. (2024). Shortening and Personalizing Psychodiagnostic Assessments with Decision Tree-Machine Learning Classifiers: An Application Example Based on the Patient Health Questionnaire-9. *International Journal of Mental Health and Addiction*. <https://doi.org/10.1007/s11469-024-01332-x>

10. Dey, S., Aziza Tahsin Nawshin, Hossain, A., & Rahman, R. M. (2022). *Machine Learning Combined with PHQ-9 for Analyzing Depression in Bangladeshi Metropolitan Areas*. <https://doi.org/10.1109/is57118.2022.10019625>

11. Kim, S., Lee, H.-K., & Lee, K. (2021). Which PHQ-9 Items Can Effectively Screen for Suicide? Machine Learning Approaches. *International Journal of Environmental Research and Public Health*, 18(7), 3339. <https://doi.org/10.3390/ijerph18073339>
12. Glavin, D., Eoin Martino Grua, Carina Akemi Nakamura, Scazufca, M., Santos, Wong, G. H. Y., Hollingworth, W., Peters, T. J., Araya, R., & Van. (2023). Patient Health Questionnaire-9 Item Pairing Predictiveness for Prescreening Depressive Symptomatology: Machine Learning Analysis. *JMIR Mental Health*, 10, e48444–e48444. <https://doi.org/10.2196/48444>
13. Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2001). The PHQ-9: Validity of a Brief Depression Severity Measure. *Journal of General Internal Medicine*, 16(9), 606–613. <https://doi.org/10.1046/j.1525-1497.2001.016009606.x>
14. Lowe, B., Unutzer, J., Callahan, C. M., Perkins, A. J., & Kroenke, K. (2004). Monitoring Depression Treatment Outcomes With the Patient Health Questionnaire-9. *Medical Care*, 42(12), 1194–1201. <https://doi.org/10.1097/00005650-200412000-00006>
15. Kroenke, K. (2021). PHQ-9: Global uptake of a depression scale. *World Psychiatry*, 20(1), 135–136. <https://doi.org/10.1002/wps.20821>
16. Inoue, T., Tanaka, T., Nakagawa, S., Nakato, Y., Kameyama, R., Boku, S., Toda, H., Kurita, T., & Koyama, T. (2012). Utility and limitations of PHQ-9 in a clinic specializing in psychiatric care. *BMC Psychiatry*, 12(1). <https://doi.org/10.1186/1471-244x-12-73>
17. Ford, J., Thomas, F., Byng, R., & McCabe, R. (2020). Use of the patient health questionnaire (PHQ-9) in practice: Interactions between patients and

physicians. *Qualitative Health Research*, 30(13), 2146–2159.
<https://doi.org/10.1177/1049732320924625>

18. Gliklich, R. E., Leavy, M. B., Cosgrove, L., Simon, G. E., Gaynes, B. N., Peterson, L. E., Olin, B., Cole, C., DePaulo, J. R., Wang, P., Crowe, C. M., Cusin, C., Nix, M., Berliner, E., & Trivedi, M. H. (2020). Harmonized Outcome Measures for Use in Depression Patient Registries and Clinical Practice. *Annals of Internal Medicine*, 172(12), 803–809. <https://doi.org/10.7326/m19-3818>

19. Ma, S., Kang, L., Guo, X., Li, H., Yao, L., Bai, H., Chen, C., Hu, M., Li, D., Du, H., Ai, C., Wang, F., Wang, G., Li, R., & Liu, Z. (2021). Discrepancies between self-rated depression and observed depression severity: The effects of personality and dysfunctional attitudes. *General Hospital Psychiatry*, 70, 25–30.
<https://doi.org/10.1016/j.genhosppsych.2020.11.016>

20. Glavin, D., Maekawa, E., Grua, E. M., Nakamura, C. A., Scazufca, M., Araya, R., Peters, T. J., Hollingworth, W., & van de Ven, P. (2022). Other PHQ-9 item pairings are better than the PHQ-2: A Machine Learning analysis. *Procedia Computer Science*, 206, 101–110. <https://doi.org/10.1016/j.procs.2022.09.089>

21. Haq, I., Nayan, M. H., Uddin, Ms. G., Hossain, M., Alam, M., Zinnia, M., Rahman, M., Ria, R., & Haq Methun, M. (2022). Comparison of the performance of machine learning-based algorithms for predicting depression and anxiety among University Students in Bangladesh: A result of the first wave of the COVID-19 pandemic. *Asian Journal of Social Health and Behavior*, 5(2), 75. https://doi.org/10.4103/shb.shb_38_22

22. Choudhary, S., Thomas, N., Ellenberger, J., Srinivasan, G., & Cohen, R. (2022). A Machine Learning Approach Detecting Digital Behavioural Patterns of Depression Using Non-intrusive Smartphone Data - A Complementary Path to PHQ-9 Assessment: A Prospective Observational Study (Preprint). *JMIR Formative Research*.
<https://doi.org/10.2196/37736>

23. Razavi, R., Gharipour, A., & Gharipour, M. (2020). Depression screening using mobile phone usage metadata: a machine learning approach. *Journal of the American Medical Informatics Association*. <https://doi.org/10.1093/jamia/ocz221>

24. Ruba Skaik. (2021). *Predicting Depression and Suicide Ideation in the Canadian Population Using Social Media Data*. <https://doi.org/10.20381/ruor-26568>

25. Kim, S., & Lee, K. (2021). Screening for Depression in Mobile Devices Using Patient Health Questionnaire-9 (PHQ-9) Data: A Diagnostic Meta-Analysis via Machine Learning Methods. *Neuropsychiatric Disease and Treatment, Volume 17*, 3415–3430.
<https://doi.org/10.2147/ndt.s339412>

26. Miraz, Md Abdullah Ibne Aziz; Mondal, Snaholata; Ahmed, Shahriar (2025), “PHQ-9 Student Depression Dataset”, Mendeley Data, V6, doi: 10.17632/kkzjk253cy.6

27. Abdullah, Md Abdullah Ibne Aziz; Ahmed, Shahriar (2026), “PHQ-9 Annotated Social Media Posts Dataset for Automated Depression Severity Assessment”, Mendeley Data, V1, doi: 10.17632/fs2r7zvmj3.1

Plagiarism Report

(Plagiarism should be less than 25%)

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